Final Contract Report

Durable Low-Cost Pressure Vessels for Bulk Hydrogen Storage

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Table of Contents

Acknowledgements 1

Disclaimers **1**

List of Figures **4**

List of Tables **7**

Abstract **8**

Executive Summary **9**

1 Introduction **11**

2 Program Results **14**

2.1 Objective 1: Selection of Cylinder Sizes and Performance Specifications **14**

2.2 Objective 2: Design Analysis **15**

2.2.1 Liner Analysis **16**

2.2.2 Assessment of Wire-Wrapped Vessel **18**

2.2.3 Estimation of Autofrettage Pressure **20**

2.2.4 Estimation of Residual Stresses Following Autofrettage **24**

2.2.5 Stresses in the Critical Regions of the Liner During Service **26**

2.3 Objective 3: Fracture Mechanics Analyses **33**

2.3.1 Environment Assisted Fatigue Crack Growth During Service Loading **33**

2.3.2 Design Life Estimation **35**

2.4 Summary of Design Analysis and Fracture Mechanics Analyses Results **41**

2.5 Objective 4: Cylinder Manufacturing **42**

2.5.1 Liner Procurement **42**

2.5.2 Wire Wrapping of the Cylinder **44**

2.5.3 Estimation of Cylinder manufacturing Costs **46**

2.6 Objective 5: Obtain ASME U3 Certification for the H_2 Storage Vessel **48**

2.6.1 Global Criteria of Loading **48**

2.6.2 Local Criteria of Loading **48**

2.6.3 Buckling and Service Criteria **49**

2.6.4 Hydrostatic Testing and Ratcheting Criteria **49**

2.6.5 Life Assessment of the Vessel **50**

2.6.5.1 Life Assessment of the Wire Wrap **50**

2.6.5.2 Life Assessment of the Liner **50**

2.7 Objective 6: Commercialization and TRL Assessment **55**

3 Summary and Conclusions **56**

4 References **57**

Appendix A: User Design Specifications **58**

Appendix B: Commercialization Plan **67**

Appendix C: Technology Maturation Plan **78**

List of Figures

Fig. 1.1 – A Schematic of the WireTough's Type II-S cylinder for ground storage of hydrogen at high pressures.

Fig. 1.2 – WireTough's cylinders can be stacked to build modular energy banks.

Fig. 2.1 –Drawing of the liner analyzed.

Fig. 2.2 – A comparison between the input stress-strain behavior for A372-Grade E Class 70 steel at 93.3 ^{0}C (200 ^{0}F) and a highly stressed element in the liner showing good agreement between the two.

Fig. 2.3 – Finite element model of the wire-wrapped pressure vessel.

Fig. 2.4– Von Mises strain (or equivalent strain) at the OD of the liner as a function of internal pressure. The autofrettage pressure is chosen when the strain equals the elastic strain plus an equivalent plastic strain of 0.3%.

Fig. 2.5 – Distribution of hoop stress across the liner wall thickness and the thickness of the wire wrap due to wrap pre-stress, at the autofrettage pressure, and after autofrettage. All liners in these analyses had an OD of 617 mm, a wall thickness of 22.9 mm, and the wrap thickness of 17.06 mm; (a) for a liner made from material with yield strength of 482.5 MPa and (b) for a liner made from a material with a yield strength of 703.5 MPa.

Fig. 2.6 – The relationship between autofrettage pressure and (a) wall thickness for two yield strengths, and (b) yield strength for a liner thickness of 22.9 mm. The liner OD was 617 mm, and the wrap thickness was 17.06 mm in both cases.

Fig. 2.7 – Autofrettage pressure as function of outside diameter, wall thickness and yield strength of the liner material.

Fig. 2.8– Comparison between predicted autofrettage pressure from equation (3) and that calculated from finite element analyses.

Fig. 2.9 – The residual stress distribution in the liner wall following autofrettage. The analyses results are for liners with an OD of 617 mm and a wrap thickness of 17.06 mm.

Fig. 2.10 –Residual stress pattern in the liner after autofrettage for cylinders of different ODs and yield strength.

Fig. 2.11 - Predicted residual circumferential stress at the ID from equation (2.4) and that calculated from the finite element analyses under similar conditions.

Fig. 2.12 - Predicted residual circumferential stress at the OD from equation (2.5) and that calculated from the finite element analyses under similar conditions.

Fig. 2.13 – Circumferential stress distribution between ID and OD of a liner with a thickness of 22.9 mm, and a material yield strength of 482.5 MPa at various operating pressures in the mid-section of the cylinder.

Fig. 2.14 – Circumferential stress distribution for an MOP of 50 MPa in the transition region from the wrap to the dome at an axial distance of 1.76 mm from the liner/wrap interface. The liners analyzed were of three different ODs and two thicknesses, and S_y=703.5 MPa.

Fig. 2.15 – Hoop stress distribution at the ID at a pressure of 41 MPa and at ambient pressure along the axial direction as a function of axial distance from the liner/wrap intersection for two wall thicknesses. The maximum stress occurs at a point approximately 1.76 mm from the wire wrap/cylinder intersection towards the dome.

Fig. 2.16 – Maximum hoop stress as function of wall thickness on the OD at a pressure of 41 MPa and at ambient pressure for a yield strength of 703.5 MPa.

Fig. 2.17 – Circumferential stress distributions from ID to OD at a pressure of 0 and 41 MPa for a 617 mm OD liner at a point 1.76 mm from the wrap/liner interface.

Fig. 2.18- Axial stress at ID and OD of a cylinder subjected to an internal pressure of 0 and 41 MPa for liner thicknesses of 22.9 and 25.2 mm for liners with a yield strength of 703.5 MPa.

Fig. 2.19 – Axial stress distribution from ID to OD at two locations along the unwrapped portion of the cylinder, at an axial distance of 1.76 and 6.76 mm. The liner OD was 617 mm.

Fig. 2.20 – The effect of OD, liner wall thickness, and the wrap thickness on the circumferential stresses in the transition region of the liner.

Fig. 2.21- Schematic representations of the FCGR behavior from the ASME Section VIII-3 master curve for A372 Grade J/E Class 70 steels as a function of load ratio, R, in high pressure gaseous hydrogen for pressures of 100 MPa.

Fig. 2.22 – The geometry of the flaw that determines the design life of the vessel.

Fig.2. 23- Definition of the angle ϕ at any point P on the boundary of an elliptical crack. The value of $\phi = 0$ on the surface and 90° at the deepest point.

Fig. 2.24 – Design life of cylinders with 617 mm OD liners for full pressure fatigue cycles as a function of MOP, initial flaw shapes and initial flaw size.

Fig. 2.25 – Design lives of cylinders with 508 mm OD liners and a wrap thickness of 17.08 mm (48 wire layers) as a function pf MOP for wall thicknesses of 22.9 and 25.2 mm, and initial flaw depth of 0.5 mm and 1 mm and a/2c values of 0.2 and 0.33.

Fig. 2.26 – Picture of the two Tenaris seamless liners being delivered to WireTough's facility.

Fig. 2.27 (Top) – Schematics of the dome regions of the liners showing points where thickness measurements were made, (bottom) results from those measurements in multiple locations.
Fig. 2.28 – Picture of a wrapped cylinder coated with a UV resistant coating ready for autofrettage and hydrotesting.

Fig. 2.29 – Cost of hydrogen storage in $ per KG of hydrogen stored using Type II-S cylinders as a function of Annual Production Volume in a single production line. The cost is compared to the cost of using Type 1 cylinders.

Fig. 2.30 – Locations of assumed flaws in the cylinder for the fatigue crack growth analyses.

Fig. 2.31 – Two finite element models used to compute stresses in the knuckle region of the liner. In the first model (a) the wall thickness is 22.9 mm throughout the dome and nozzle region and (b) the thickness profile in the dome region as dictated by intersection of the extensions of the inner surface of the liner in the nozzle region and the curvature of the dome at the ID. The wall thicknesses in the various locations are shown in this figure in inches and is as high as 38 mm (1 inch = 25.4 mm).

Fig. 2.32 -Estimated stress distributions at location 5 in the knuckle region at maximum and minimum operating pressures using Model 1 that assumes the wall thickness to be the same in the dome and knuckle regions (Fig. 2.31a) and Model 2 using variable wall thickness (Fig. 2.31b) showing significant reduction in stress when Model 2 is used.

List of Tables

Table 2.1 – Liner choices available and their dimensions and storage capacities. MOP and MWAP are the maximum operating pressure and the Maximum allowable working pressure, respectively of the wrapped cylinder. OD = outside diameter of the liner.

Table 2.2 – Wrapped cylinder characteristics utilizing the liners listed in Table 2.1.

Table 2.3 – Matrix of cases analyzed using finite element analyses.

Table 2.4 – Summary of l stresses in the critical region in the pressurized cylinder

Table 2.5 – Constants in equation (2.6) for describing the environment assisted fatigue crack growth behavior in SA372 Grade E/J Class 70 steels in high pressure hydrogen (p = 102 MPa).

Table 2.6 – Design lives resulting from the parametric analyses that varied MOP, wall thickness, initial flaw size, and initial flaw shape. The liner OD in all these cases was 617 mm with a wrap thickness of 17.06mm.

Table 2.8 – Chemistry of the liner steel and the acceptable composition range (wt %).

Table 2.9 – Tensile test results of the liner material.

Table 2.10 – Calculated autofrettage pressures and hydrotest pressures for the two cylinders built.

Table 2.11 – List of materials of construction for manufacturing Type II-S wire wrapped vessel.

Table 2.11 – List of materials of construction for manufacturing Type II-S wire wrapped vessel.

Table 2.12 - Cost and weight savings in switching from Type 1 to Type II-S cylinders for MOP of 350, 500,

Table 2.13 – Crack models used in the fracture mechanics analysis for estimating design life at various points of the cylinder.

Table 2.14 shows the predicted design life/ inspection intervals that were estimated from these calculations.

Abstract

Long-duration energy storage (LDES) in the form of gaseous hydrogen is vital for fossil fuel power plants to reduce frequent shutdowns and rapid start-ups. It's also crucial in renewable energy power plants for grid stabilization. WireTough Cylinders (WTC) has developed a Type II-S pressure vessel that uses steel liners that are hoop-wrapped with multiple layers of high-strength, cost-effective, and readily available steel wires. This design enables the vessel to withstand significantly higher pressures. It addresses hydrogen embrittlement issues and cyclic loading from repeated pressurization and depressurization during service. The WireTough design is cost-efficient, robust, and fatigue-resistant (with a 30-year design life) and is ASME certified for ground storage of hydrogen at a maximum operating pressure (MOP) of 350 bar and a design pressure of 500 bar. The cylinder has a 645 mm outside diameter and a water capacity of 1,715 liters, capable of storing 40.7 KG of hydrogen at MOP. Two prototypes were successfully produced to showcase the manufacturing technologies, with one being certified under ASME Section VIII Division 3 rules. Preliminary design work for a similar Type II-S vessel with a 500 bar MOP was also completed. The 350 bar and 500 bar Type II-S vessels were shown respectively to be 47.5% to 60% more cost-effective and 55% and 62.5% lighter based on per KG of hydrogen stored, when compared with their Type 1 counterparts.

Executive Summary

Fossil-fuel electricity generating plants rely on capital-intensive equipment that must frequently be shut down and rapidly ramped up to accommodate varying energy demands throughout the day. The resulting thermal cycling causes damage to plant components, leading to costly, unplanned outages and inefficient energy conversion. This problem underscores the importance of integrating environmentally friendly on-site long duration energy storage (LDES) facility within the plant. LDES facilities are also needed for balancing the power grid in renewable energy power plants and at fueling stations serving various transportation needs. Storing gaseous hydrogen at high pressures (35-50 MPa) in specially designed pressure vessels is an attractive LDES solution.

WireTough Cylinders (WTC) has developed an innovative design and manufacturing technologies for cost-effective, durable, fatigue and hydrogen-embrittlement resistant, ASME-Section VIII-Division 3 certified cylinders for storing hydrogen at high pressures. These cylinders feature a seamless steel liner encased in a steel wire jacket composed of multiple layers of ultra-high-strength steel wires with tensile strength exceeding 2.0 GPa. These wires are meticulously wound around the liner's axial length and secured with epoxy. After wire wrapping, the cylinders undergo autofrettage, a process that imparts high compressive stresses throughout the liner wall at ambient pressure. This significantly enhances the fatigue life of the cylinders and strengthens the bond at the interface of the steel wire jacket and the liner.

The product lineup comprises three models of all-steel Type II-S cylinders, each designed for specific pressure ratings. These models store 40.7 KG of hydrogen at 350 bars, 42 KG at 500 bars, and 29.5 KG at 700 bars in a single vessel. The 350-bar cylinder has an outer diameter of 645 mm, measures 7600 mm in length, and has a capacity of 1715 liters. This design is fully certified. The 500-bar cylinder features an outer diameter of 543 mm, a length of 9000 mm, and a capacity of 1400 liters, while the 700-bar cylinder has an outer diameter of 441 mm, a length of 9000 mm, and a capacity of 765 liters. The latter two designs are ready for certification but have not yet gone through the certification steps.

In-depth finite element analyses were conducted on both the steel liner and the finished pressure vessels, ensuring compliance with the relevant sections of the ASME code. These analyses encompassed stresses originating from the wire-winding process, residual stresses from autofrettage, and subsequent service loading. Fracture mechanics analyses, in line with ASME Section VIII Division 3 Codes, were employed to estimate the cyclic life of the cylinder. Assessments involved postulating surface flaws measuring 1 mm in depth and 5 mm in length at various locations within the liner to assess the vessel's integrity.

The 1715 L capacity liners for producing cylinders with a 350-bar operating pressure rating were obtained from Tenaris, USA. These liners fully adhered to the design specifications to produce full-size prototypes, which effectively demonstrated the manufacturing technology. Two cylinders underwent all steps of wire wrapping. One of

these wrapped cylinders was also subjected to autofrettage and hydrotesting and was certified by an ASME authorized inspector, affirming its compliance with all design requirements.

The process also included the formulation of a User Design Specification (UDS) to clearly define the cylinder's capabilities for service. Subsequently, a Manufacturer's Design Report (MDR), developed by Structural Integrity Associates, was produced to fulfill the UDS requirements. This report evaluated the WTC design against all ASME Section VIII-Div 3 codes, confirming full compliance.

The MDR revealed that the knuckle region on the inner surface at the intersection between the nozzle and the dome is the critical, life-limiting region of the liner. In this region, liner wall thicknesses were found to be more than 60% greater than initially assumed in stress estimations. Factoring in these realistic thicknesses, the vessel's design life was estimated at 6,000 cycles, equivalent to 30 years at a rate of 200 cycles per year. Design lives in other areas were higher. Also, the critical region is not covered by the wire jacket and therefore accessible for inspection while in service to potentially extend the cylinder life beyond the 30 years of design life.

The manufacturing costs for the 1715 L, 350-bar cylinder were estimated. The costs were adversely affected by the doubling of steel prices and increased shipping costs during the COVID-19 pandemic. In scenarios where fewer than 20 cylinders are produced annually, manufacturing costs were estimated at $1478 per kilogram of stored hydrogen. However, this cost significantly reduces to $827 per kilogram of hydrogen stored—a 44% reduction—when annual production volume (APV) increases to 200. A single manufacturing line operating in two shifts can achieve an annual production of 200 cylinders, so this cost is achievable. These costs surpass the goals set by the Department of Energy (DOE) for bulk hydrogen storage.

The technology readiness level (TRL) for this innovation was raised from TRL-4 to TRL-8 during the program. The 350-bar, 1715 L hydrogen storage cylinder is now eligible for commercial deployment, and the 500-bar and 700-bar cylinders can be brought to the commercial market within six-months to one year.

1. Introduction

Capital intensive equipment in fossil-fuel electricity generating plants is required to shut-down frequently and ramp-up rapidly to manage variable energy demands within a single day. Damage to plant equipment/components due to such thermal cycling causes unplanned outages that can be expensive and results in inefficient energy conversion. Integrating an onsite environment-friendly long term energy storage (LDES) facility addresses this challenge. Utilizing excess electricity during low demand periods in fossil-fuel plants or during peak generation periods in renewable energy plants to generate hydrogen and storing it at high pressures of 350-500 bar is attractive. Advances in pressure vessel technology have made this energy storage approach technically viable and cost-effective on a large scale [1]. Other approaches include liquification of hydrogen, but that requires large amounts of energy to initially cool the gas and then to maintain the low temperatures during storage. Metal hydrides require considerable heat to discharge the stored hydrogen and the process is slow. Surface hydrogen storage systems such as C-nanotubes are currently not sufficiently mature [2]. The compressed H_2 approach to LDES also competes favorably with non-hydrogen approaches such as batteries that degrade over time, retain energy only for short duration, and create chemical waste at the end of life. The pumped water approach requires large amounts of land, and thermal energy storage systems are currently not efficient.

WireTough Cylinders (WTC) has developed the design and manufacturing technologies for *low-cost, durable, and fatigue damage-resistant* ASME BPVC Section VIII-Division 3 certified cylinders to store hydrogen at high pressures. WireTough's solution consists of a patented design, and proprietary manufacturing processes utilizing a seamless steel liner and a steel wire jacket made from multiple layers of ultra-high strength (> 2.0 GPa), automotive tire chord quality steel wires that are wrapped around the axial length of the liner and held together with an epoxy. A schematic drawing of WireTough's Type II-S design is shown in Fig. 1.1. The wire jacket carries a significant portion of the pressure load. After wire wrapping, the cylinders are subjected to an autofrettage process which locks high compressive stresses throughout the thickness of the liner wall. The process significantly improves the fatigue life of the cylinders and simultaneously strengthens the interface bond between the steel wire jacket and the liner that prevents moisture from seeping in during service. Efficient cylinders can be designed to operate at maximum operating pressures in the range of 350 to 900 bar.

One KG of hydrogen contains the heat equivalent of 39 kWh of electricity and assuming an efficiency of 60% for fuel cells, it will yield back an estimated 23.4 kWh of electricity. A 1715-liter cylinder that has an OD of 645 mm and a length of 7600 mm and can safely withstand a hydrogen pressure that cycles between 350 and 35 bar can store about 40.7 KG of hydrogen that can deliver approximately 950 KWh of electricity. A modular design will permit building energy banks of different capacities. For example, a six-cylinder stack (Fig. 1.2) can build an energy bank that can store 5.4 MWh of power in a module with a footprint of less than 200 Sq. ft.

WireTough initiated a program in March of 2020 to develop an ASME certified hydrogen storage cylinder. WireTough partnered with Siemens Energy Technology Center in Charlotte, NC to conduct the program. The objectives of this DOE cooperative research program were as follows:

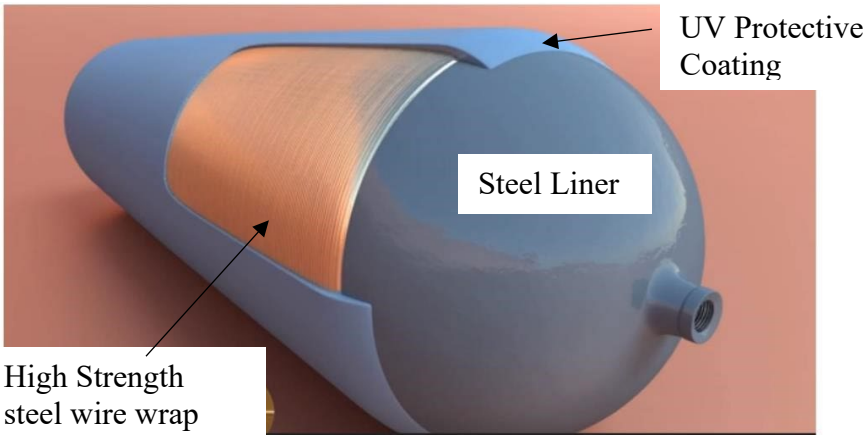


Fig. 1.1 – A Schematic of the WireTough's Type II-S cylinder for ground storage of hydrogen at high pressures.



Fig. 1.2 – WireTough's cylinders can be stacked to build modular energy banks.

Objective 1: To design and optimize a high strength wire-wrapped all steel Type II-S cylinder with a capacity to store between 40 to 50 KG of hydrogen at a pressure of 350 -

500 bar (5,000 to 7,250 psi). The design must be fully compliant with the ASME Section VIII-Division 3 design codes for storing hydrogen using only code approved materials. The bare cylinder weight will demonstrably be lighter and cost-effective in comparison to Type 1 cylinders that are currently in use.

Objective 2: To conduct finite element analyses of (a) the steel liner by itself and (b) the finished pressure vessel to determine the magnitude and distribution of stresses during manufacturing and during service to ensure that the design is safe, and it can be economically produced.

Objective 3: Perform Fracture Mechanics analyses in accordance with the ASME Section VIII Division 3 rules to estimate the cyclic life of the cylinder.

Objective 4: To procure liners that are consistent with design specifications and produce full-size prototypes of the vessel to demonstrate the manufacturing technology. The cost target was $900 (based on pre-COVID costs of steel) or less per KG of stored hydrogen with the potential to further reduce it by 25%, when produced in a single integrated facility in large quantities. Demonstrate that making these cylinders requires only commonly available construction materials.

Objective 5: Produce a User Design Specification (UDS) and a Manufacturing Design Report (MDR) to pave the way for ASME U3 Stamping of the vessel.

Objective 6: The estimated technology level at the start of the program was TRL 4 that will be pushed to at least TRL 5 or higher at the conclusion of the program.

These objectives were to be met by executing 7 tasks within the program. The next section describes the results from the program that are organized to show how each of the above objectives was met.

2. Program Results

The program results described in this section are aligned with the objectives of the program stated above.

2.1 Objective 1: Selection of Cylinder Sizes and Performance Specifications

The cylinder specifications were selected after consideration of several factors including manufacturability, maintainability during service, cost, and durability. These are as follows:

- Maximum operating pressures (MOP) of 350, 500, and 700 bar based on considerations of volumetric energy density of hydrogen and maximizing cylinder capacity at those pressures.

- Choice of liners made from commonly available pipe sizes with outside diameters of 610, 508, and 406 mm.

- Stress transfer ratio from the liner wall to the wire jacket to optimize autofrettage pressures necessary for optimum fatigue durability during service and reducing risk of failure in the dome region of the vessel during autofrettage.

- The maximum length of the pressure vessel to be 10,000 mm or less based on limits of transportation, maximum finished weight of the vessel of less than 4,500 KG due to winding machine limitations and the capacity of the material handling systems.

- Amount of hydrogen stored at the maximum operating pressure (MOP) of 40 – 60 KG.

- Number of wire layers in the steel wire wrap (40 to 48 layers).

- Use of only ASME Code approved materials for pressure vessels for storing hydrogen,

- Even though the vessel weight expressed in Kilogram (KG) per KG of H_2 stored is not a primary consideration for ground storage, the cost, which is a primary consideration, is directly linked to the vessel weight.

These considerations narrowed the choices of liners listed in Table 2.1 and wrapped cylinder dimensions listed in Table 2.2. Clearly, the ideal pressures for bulk storage are between 350 to 500 bar from the weight per KG of hydrogen stored and the efficiencies obtained in transferring stresses on to the high strength wire jacket. Further, a 610 mm OD liner suitable for producing a wrapped vessel capable of storing 40.7 KG of H_2 at 350 bar pressure is the optimum among the three. Thus, the design analysis in the support of Objective 2 described in the next section will focus on that vessel size.

Table 2.1 – Liner choices available and their dimensions and storage capacities. MOP and MWAP are the maximum operating pressure and the Maximum allowable working pressure, respectively of the wrapped cylinder. OD = outside diameter of the liner.

MOP (bar)	MAWP (bar)	Liner OD (mm)	Min Wall Thickness (mm)	Length (mm)	Water Volume, (L)		Weight, (KG) min	Weight of H_2 stored at MOP (KG)
					Max	Nominal		
350	500	610	22.9	7600	1761	1715	2339	40.7
500	690	508	25.4	9000	1392	1350	2564	42.0
700	1000	406	31.75	9000	785	765	2494	29.5

Table 2.2 – Wrapped cylinder characteristics utilizing the liners listed in Table 2.1

MOP (Bar)	Cylinder OD (mm)	Weight of the Wrap, KG	Length (mm)	Number of Wire Layers	Cylinder Weight, KG	Cylinder Weight/KG of Stored H_2 (KG)	Stress Transfer Ratio
350	645	1343	7600	48	3682	90.3	0.277
500	543	1370	9000	48	3974	93.5	0.247
700	441	1110	9000	48	3604	122.15	0.191

2.2 Objective 2: Design Analysis

Finite element analyses were conducted (a) of the steel liner by itself (b) of the wire wrapped pressure vessel to determine the magnitude of residual stresses created during wire wrapping and during autofrettage, and (c) of the finished vessel subject to service loading. Fracture mechanics-based methods that are part of the ASME Section VIII Division 3 Codes were used to estimate the cyclic life of the cylinder. The results from these analyses are described in the following sections.

The finite element analyses were conducted by Siemens using Siemens' Simcenter 3D. The simulation workflow was automated using NX Open API for each design, including (i) design update, (ii) setting the appropriate initial and boundary conditions, (iii) solve the Simcenter Nastran simulations, and (iv) results extraction and analysis after each simulation. The design space exploration was performed with Siemens Simcenter HEEDS that accelerated the design space exploration by enabling automation of the simulation analysis workflows and enabling distributed execution using available computational resources.

2.2.1 Liner Analyses

The liner chosen is shown in Fig. 2.1 and was virtually pressurized and the stress and strain at various points were estimated. The purpose was to estimate the burst pressure of the proposed liner. The liner material was SA372 grade E or J steel and the yield strength and the ultimate strength chosen for the analyses were the minimum values specified for this grade of material. These values were 482.5 MPa and 827 MPa, respectively. The OD for determining the burst pressure was 617 which is the nominal OD +1% allowable variation within the liner specifications. The thickness was 22.9 mm which is the minimum specified thickness. Figure 2.2 shows that an excellent comparison was obtained between the predicted stress-strain behavior at a highly stressed point in the liner with the input stress-strain for the liner material. This result validates the accuracy of the finite element results.

The predicted burst pressure for the liner by itself was 72 MPa from the above analysis. Structural Integrity Associates have also conducted an elastic-plastic analysis of the same liner and have estimated the burst pressure as 64 MPa using a more conservative failure criterion based on onset of plastic instability. A third estimate of the burst pressure, based on the well-known Faupel formula, equation (2.1) yielded the most conservative value of 60.78 MPa. This value was used to establish the design pressure of the vessel.

$$P_b = \frac{2}{\sqrt{3}} S_y \left(2 - \frac{S_y}{S_u}\right) ln \frac{D_o}{D_i} \quad (2.1)$$

Where, P_b = Burst Pressure, S_y = minimum 0.2% yield strength, S_u= minimum ultimate tensile strength, D_o = outside diameter of the liner, and D_i = inside diameter of the liner. Table 2.1 lists the calculated maximum allowable pressures (MAWP) and the maximum operating pressures (MOP) for the various size liners from the three methods. Since equation (2.1) provides the most conservative burst pressure, it was chosen for the parametric analysis in which the wall thickness and the OD of the liners were varied. The results are shown in Table 2.1. The liner ODs range in the interval of $617 \leq D \leq 406$ mm and wall thicknesses in the interval $22.9 \leq t \leq 25.2$ mm. Depending on the OD and wall thickness, t, of the chosen liner, the MWAP can range from $50.6 \leq MAWP \leq 85.1$ MPa. The corresponding recommended maximum operating pressures (MOP) are 70% of the design pressure to allow for pressure increases due to heat during fills. MOPs are also chosen based on the desired design life of the vessels during service. This will be discussed later in the report.

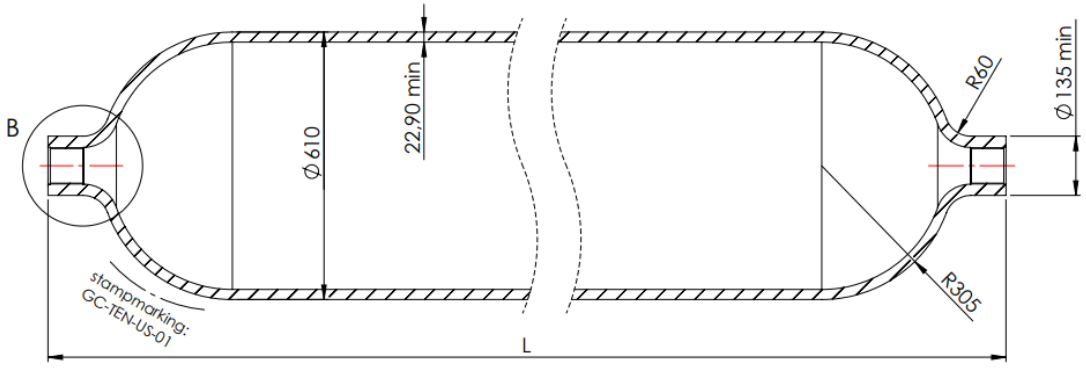


Fig. 2.1 –Drawing of the liner analyzed

Table 2.1 – Burst Pressure estimates of liners of various sizes considered

Nominal OD (mm)	Minimum Wall thickness, mm	Estimated Burst Pressure, MPa			MAWP = (P_b/1.2) MPa	Recommended MOP = 0.7(MAWP)
		FE Analysis- Siemens	FE Analysis - SIA	Faupel Formula		
617	22.9	72	64	60.78	50.65	35.7
617	25.2	-	-	68.1	56.8	37.5
508	22.9	-	-	74.4	62	43.4
508	25.2	-	-	82.4	68.7	47.6
406	22.9	-	-	94.2	78.5	54.6
406	25.2	-	-	102.14	85.1	59.6

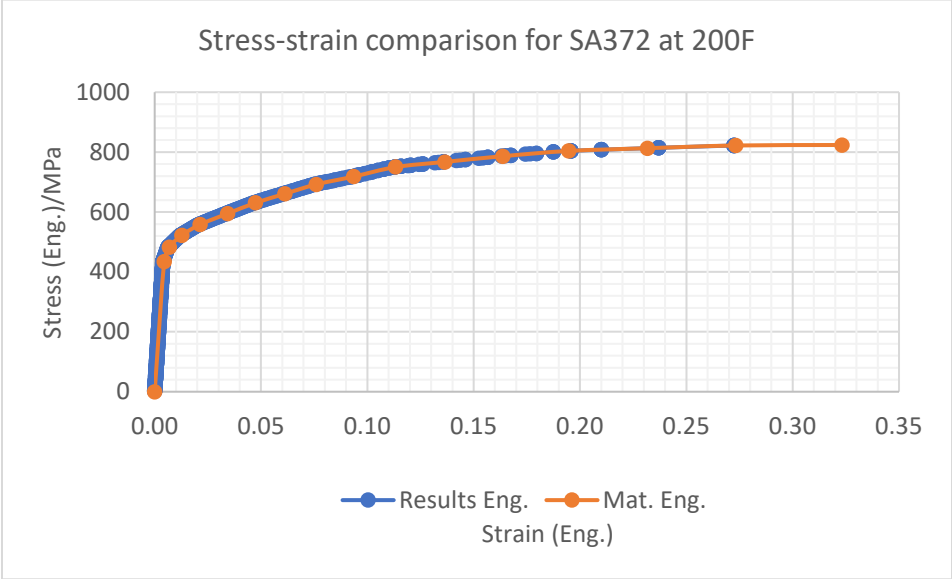


Fig. 2.2 – A comparison between the input stress-strain behavior for A372-Grade E Class 70 steel at 93.3 ^{0}C (200 ^{0}F) and a highly stressed element in the liner showing good agreement between the two.

2.2.2 Assessment of the Wire Wrapped Vessel

The entire pressure vessel assembly was evaluated under loading consisting of (1) stresses due to the wrapping process (2) residual stresses due to autofrettage and (3) stresses during service loading. The pressure vessel assembly included the liner made from SA 372 Grade E/J Class 70 steel with an OD of 610 + 0.01% = 617 mm. The wire jacket consisted of 48 and 42 layers of SA905 high strength wire. The wire was applied as 8 or 7 bundles with each bundle consisting of six wire layers. The first bundle was offset 9 mm from the start of the dome and each subsequent layer was offset 7 mm from the layer beneath it. Figure 2.3 shows the finite element model of the wire-wrapped vessel.

Several finite element analyses were conducted in the range of wall thickness, t , $22.9 \leq t \leq 25.2$ mm and the yield strength, Sy, $482.5 \leq S_y \leq 703.5$ MPa and for outer diameters of 406, 508, and 617 mm. The specific cases for which the finite element analyses were conducted are summarized in Table 2.3.

Table 2.3 – Matrix of cases analyzed using finite element analyses.

OD, mm	S_y, MPa	Wall Thickness, mm					Number of Wire Layers
617		22.9	23.5	24.0	24.5	25.2	48
	482.5	x	x	x	x	x	
	517	x					
	552	x					
	587	x					
	622	x					
	658	x					
	703.5	x	x	x	x	x	
617	703.5	x				x	42
508	703.5	x				x	48
508	703.5	x				x	42
406	703.5	x				x	48
406	703.5	x				x	14.22

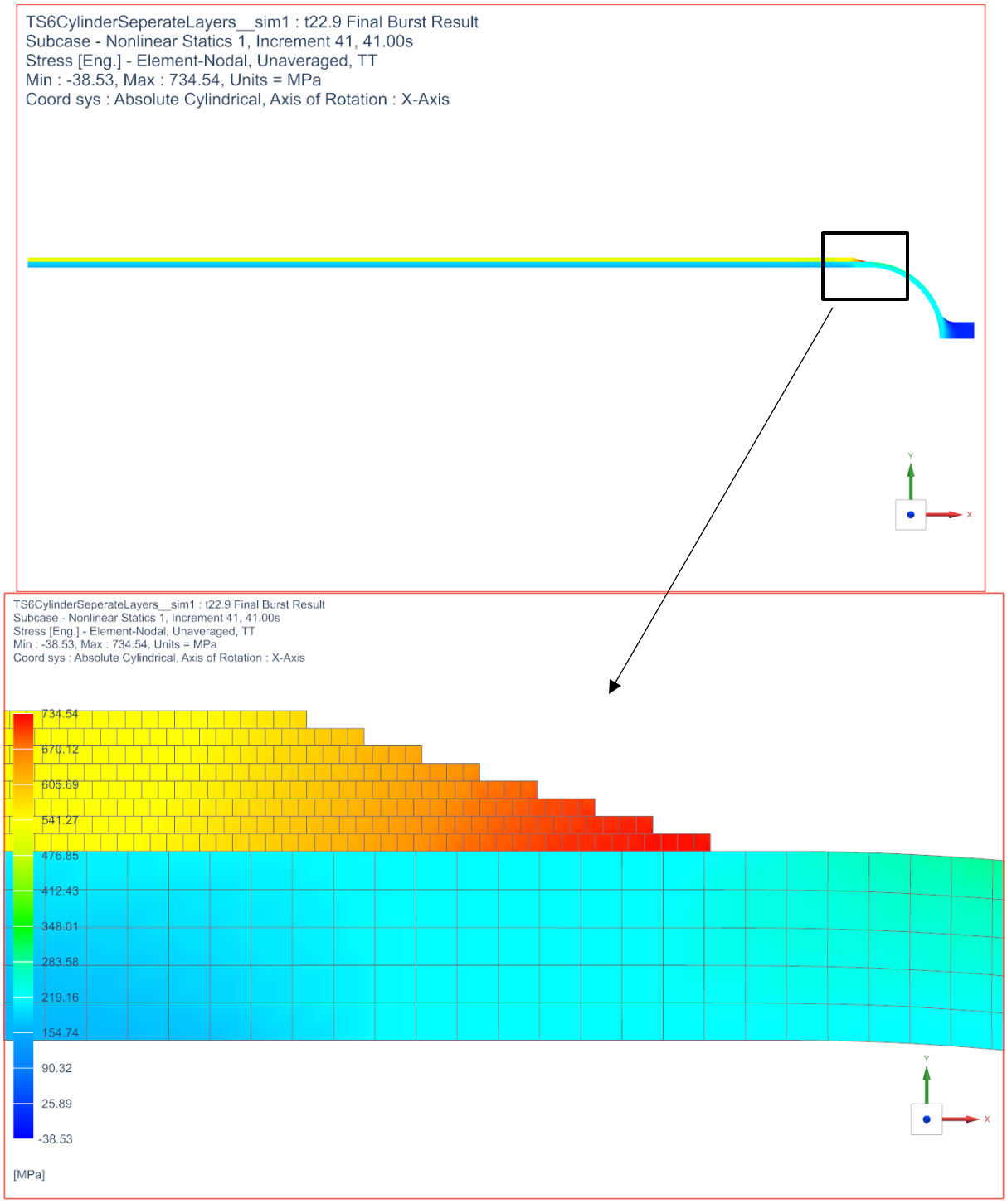


Fig. 2.3 – Finite element model of the wire-wrapped pressure vessel.

2.2.3 Estimation of the Autofrettage Pressure

The process of autofrettage is used in the pressure vessel industry to enhance the fatigue life of vessels for storing hydrogen at high pressures. As the storage pressures fluctuate during service, the amplitudes of circumferential fatigue stresses also change, affecting the fatigue life of the vessel. Autofrettage locks high compressive stresses in the entire liner wall of the vessel that decreases the magnitude of the maximum and the minimum stresses during the service cycle. This significantly reduces the environment assisted fatigue crack growth rates that depend on the maximum stress and the stress ratio, enhancing the cyclic life of the vessel.

A model to correctly estimate the autofrettage pressure necessary to introduce a repeatable and consistent residual stress pattern in the liner wall from its ID to OD in Type II-S vessels was developed. The results from cases listed in Table 2.2 were used to derive an equation for estimating the autofrettage pressure as a function of the liner wall thickness, liner material yield strength, the OD, and the wrap thickness.

Figure 2.4 shows a plot of Von Mises' equivalent strain on the outer diameter, OD, of the liner as a function of the applied internal pressure in the wire-wrapped cylinder. The autofrettage pressure is assumed to have reached when the equivalent strain on the OD of the liner is equal to the elastic strain for the stress level applied plus a plastic strain of 0.3%, of which 0.2% is associated with reaching plastic yielding plus 0.1% beyond that. This ensures that the entire liner wall deforms plastically during autofrettage, so when the pressure is removed, high compressive stresses are locked through the entire cross-section of the liner. The chosen autofrettage pressure is marked in Fig. 2.4.

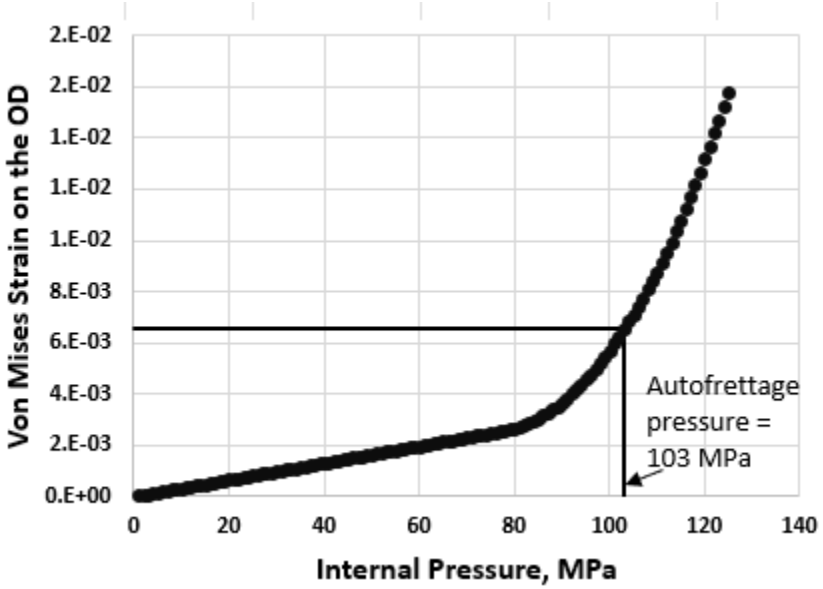


Fig. 2.4– Von Mises strain (or equivalent strain) at the OD of the liner as a function of internal pressure. The autofrettage pressure is chosen when the strain equals the elastic strain plus an equivalent plastic strain of 0.3%.

Figure 2.5 shows the typical distribution of hoop stress across the cylinder wall (a) after wrapping (b) at the autofrettage pressure, and (c) after removal of the autofrettage pressure in cylinder. The results shown are for wrap thickness of 17.06 mm (48 wire layers) and a liner wall thickness of 22.9 mm. Figure 2.5a is for a yield strength of 482.5 MPa and Fig. 2.5b for a yield strength of 703 MPa. The residual stresses due to autofrettage in the higher yield strength liner are significantly greater than in the liner with lower yield strength. As expected, the compressive residual stresses in the liner are equilibrated by tensile stresses in the wrap. At the autofrettage pressure, note that the tensile stress in the wrap is less than 825 MPa in Fig. 2.5b which is much less than the strength of the wire, conservatively specified in the ASME Code as 2 GPa. Therefore, the wrap is not at risk of fracturing during autofrettage. Also note that the interface between the wrap and the liner is expected to be under high contact pressures which is beneficial for the integrity of the cylinder.

Figure 2.6a shows the calculated autofrettage pressure as a function of the wall thickness of the liner material for two levels of yield strength, and Fig. 2.6b shows the relationship between the autofrettage pressure and the yield strength for a liner with a fixed thickness of 22.9 mm. In both cases the liner OD was 617 mm, and the wrap thickness was 17.06 mm. The autofrettage pressure increases linearly with liner thickness and with material yield strength.

Fig. 2.7 shows how the autofrettage pressure varies with the liner OD for two yield strength levels and two wall thicknesses. Again, the relationship with OD appears to be linear in all cases. Based on the observations from the results of the finite element analyses, the following non-dimensional, functional relationship is derived between the autofrettage pressure and various variables:

$$\frac{P_A}{P_{A0}} = f\left[\frac{S_y}{S_{yo}}, \frac{t}{t_o}, \frac{D}{D_o}, \frac{t_w}{t_{w0}}\right] \tag{2.2}$$

In the above equation, P_A = autofrettage pressure, P_{Ao} = autofrettage pressure for the reference thickness, t_0 = 22.9 mm, the reference yield strength of the liner material, S_{yo} = 482.5 MPa, reference OD of D_o= 617 mm, and a reference wrap thickness, t_{wo} of 17.06 mm. S_y = yield strength of the liner material, t = the liner thickness, and D = OD, and t_w = wrap thickness.

From the finite element results, an equation of the following form was derived [3] to collectively represent the effects of yield strength, wall thickness, and OD on the autofrettage pressure for a wrap thickness of 17.06 mm.

$$\frac{P_A}{P_{A0}} = \left(0.26 + 0.74\frac{S_y}{S_{y0}}\right)\left(.206 + .794\left(\frac{t}{t_0}\right)\right)\left(2.4 - 1.4\frac{D}{D_0}\right) \tag{2.3}$$

Where, $P_{A0} = 77\ MPa,\ S_{y0} = 482.5,\ D_0 = 617\ mm,\ t_0 = 22.9\ mm, t_{wo} = 17.06\ mm$

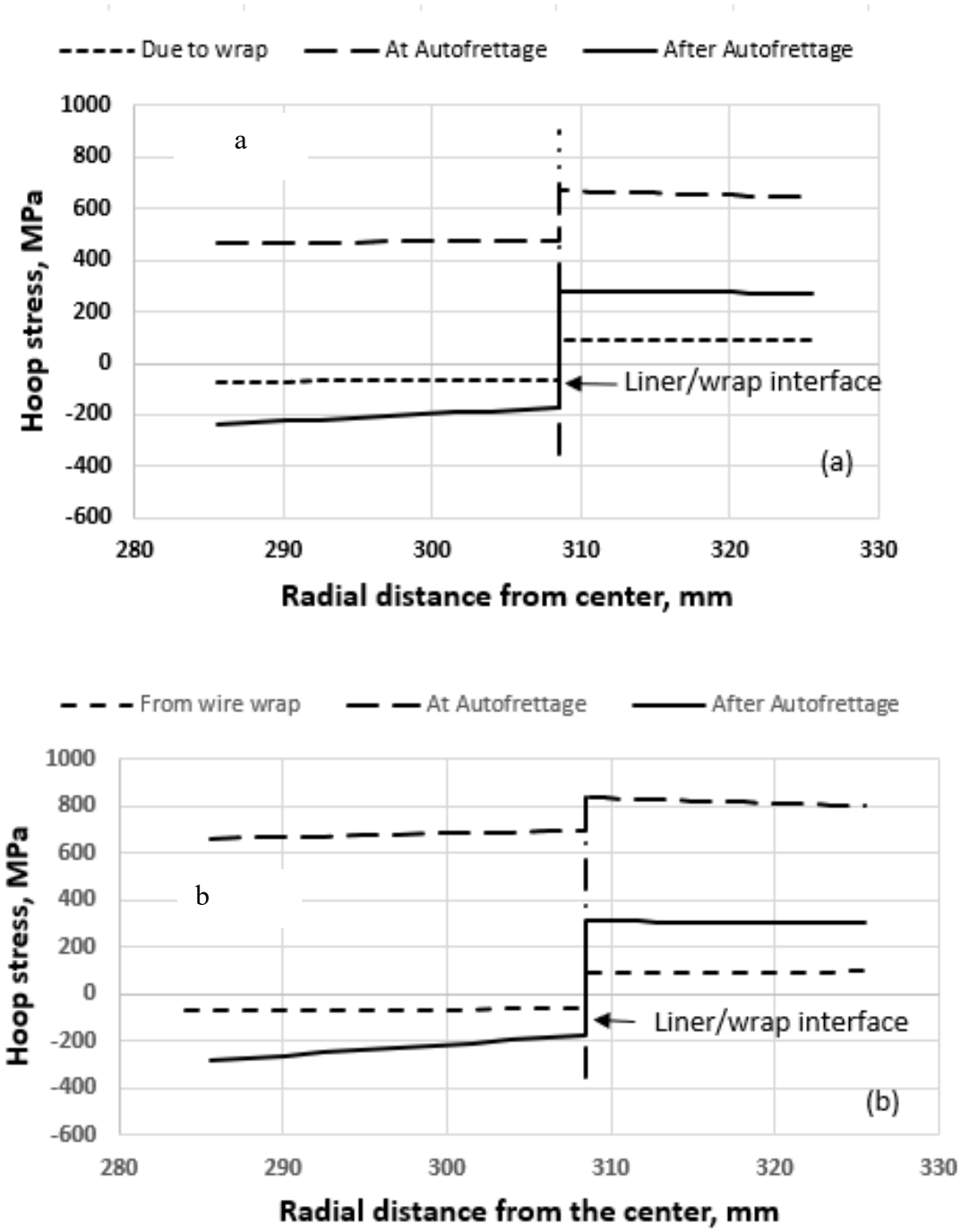


Fig. 2.5 – Distribution of hoop stress across the liner wall thickness and the thickness of the wire wrap due to wrap pre-stress, at the autofrettage pressure, and after autofrettage. All liners in these analyses had an OD of 617 mm, a wall thickness of 22.9 mm, and the wrap thickness of 17.06 mm; (a) for a liner made from material with yield strength of 482.5 MPa and (b) for a liner made from a material with a yield strength of 703.5 MPa.

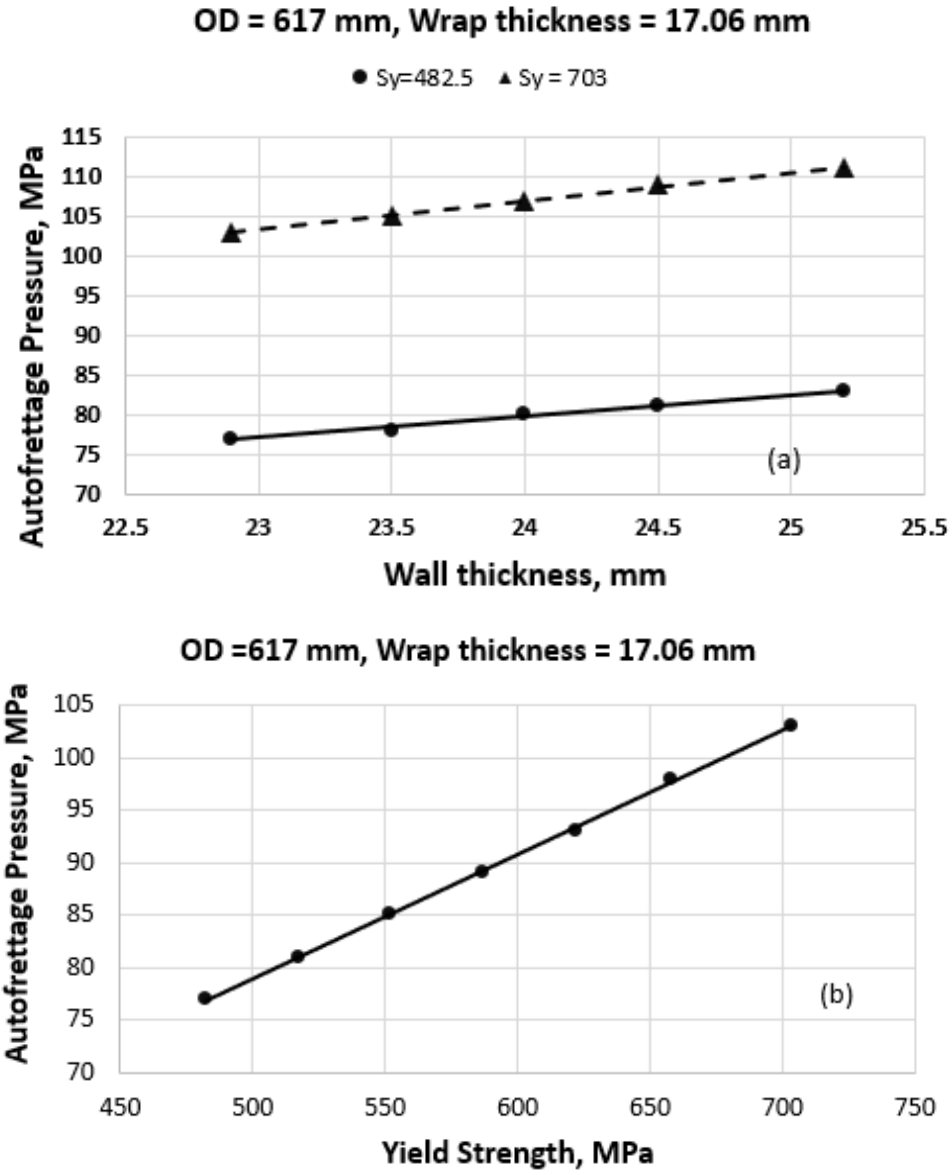


Fig. 2.6 – The relationship between autofrettage pressure and (a) wall thickness for two yield strengths, and (b) yield strength for a liner thickness of 22.9 mm. The liner OD was 617 mm, and the wrap thickness was 17.06 mm in both cases.

Figure 2.8 shows a comparison between the calculated autofrettage pressure from finite element analyses and that from equation (3) for all conditions analyzed. The agreement is excellent demonstrating that the effects of these variables are separable as assumed in equation (2.3) with each showing a linear relationship with the autofrettage pressure when referenced to their respective reference values. The bounds for which equation (2.3) holds are as follows. All length dimensions are in mm and strength is in MPa.

$22.9 \leq t \leq 25.2, \qquad 617 \leq D \leq 406, \qquad 482.5 \leq S_y \leq 703.5, \qquad 14.22 \leq t_w \leq 17.06$

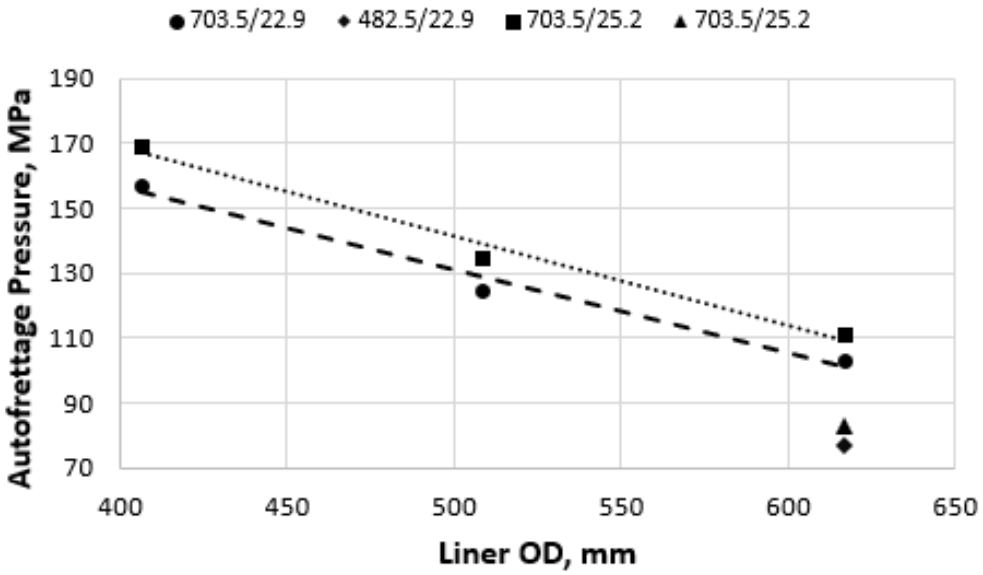


Fig. 2.7 – Autofrettage pressure as function of outside diameter, wall thickness and yield strength of the liner material.

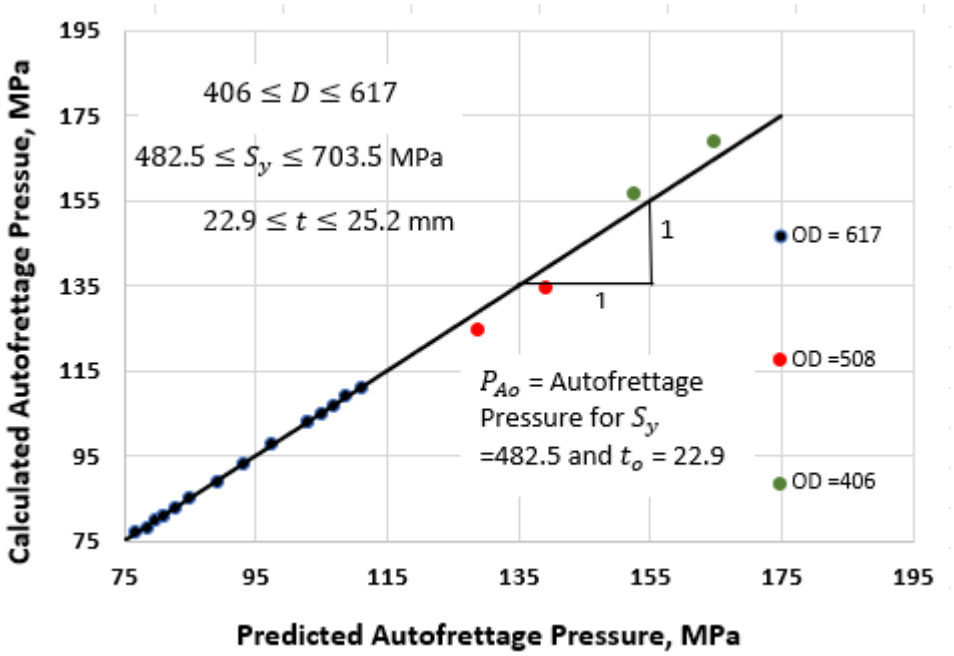


Fig. 2.8– Comparison between predicted autofrettage pressure from equation (3) and that calculated from finite element analyses.

2.2.4 Estimation of Residual Stresses Following Autofrettage

The objective of the autofrettage process is to develop a pattern of predictable compressive residual stresses in the wall of the liner that reduces the maximum stresses during service loading cycle. Since negative stresses do not contribute toward the

magnitude of ΔK during fatigue crack growth analysis, its magnitude decreases also. Figure 2.9 shows the typical residual stress distribution following autofrettage for liners of different thicknesses and yield strength. All liners included in the figure had an OD of 617 mm that were wrapped in a jacket that was 17.06 mm thick. The compressive residual stresses are the highest in magnitude on the ID and lowest on the OD and are distributed linearly between the ID and OD, that is expected for thin-wall liners. Since the autofrettage pressure is linked to a fixed amount of plastic deformation on the OD, the residual stress pattern for different thicknesses lies along the same trend. Thus, the magnitude of residual stresses in the interior of the liner wall do not vary with liner thickness but are significantly affected by the yield strength.

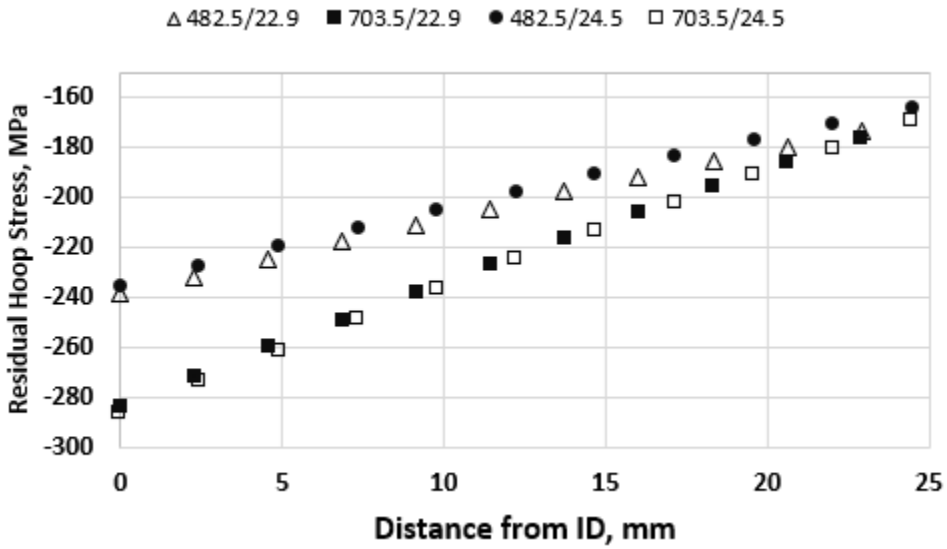


Fig. 2.9 – The residual stress distribution in the liner wall following autofrettage. The analyses results are for liners with an OD of 617 mm and a wrap thickness of 17.06 mm.

Figure 2.10 shows the variation in residual stress distribution for cylinders from liners with different ODs and yield strengths for a constant wrap thickness of 17.06 mm. The distribution of stress between ID and OD is assumed to be linear consistent with the observation from Fig. 2.9. The residual stress depends significantly on the OD of the liner and the observation that the stress pattern is similar for liners with different thicknesses also holds for the ODs of 508 mm and 406 mm.

Equations (2.4) and (2.5) represent the residual compressive circumferential stress in the liner wall at the ID and OD of the liner in its central region. These equations account for the variation in residual stress with yield strength, OD, and wrap thickness. Between the ID and OD, the stress distribution is expected to be approximately linear. As shown before, the ID residual stresses are independent of liner thickness, but the OD residual stresses do change linearly with the liner wall thickness but predictably from equation (2.4).

$$\sigma_{\theta ID} = \sigma_{\theta IDo}\left(0.6 + 0.4\left(\frac{S_y}{S_{yo}}\right)\right)\left(1.4 - .4\frac{D}{D_0}\right) \quad (2.4)$$

$$\sigma_{\theta OD} = \sigma_{\theta ID} - \left(-195 + 105\left(\frac{D}{D_0}\right)\right)\frac{S_y}{S_{yo}}\frac{t}{t_0} \quad (2.5)$$

$\sigma_{\theta ID}$ = circumferential stress at the ID of the liner, $\sigma_{\theta IDo}$= is the circumferential stress at the ID for the reference liner with $\sigma_{\theta IDo} = 238.84,\ S_{y0} = 482.5,\ D_0 = 617\ mm.\ t_0 = 22.9\ mm, t_{wo} = 17.06\ mm$. Figures 2.11 and 2.12 show a direct comparison between the predicted residual circumferential stress at the ID and OD for all cases analyzed by finite element analyses, showing very good agreement between values from equations (2.4) and (2.5) and the finite element analyses. In these figures, the lines indicating an error band of $\pm$5% are plotted to demonstrate that equations (2.4) and (2.5) have no more than 5% difference from the values determined from finite element analyses.

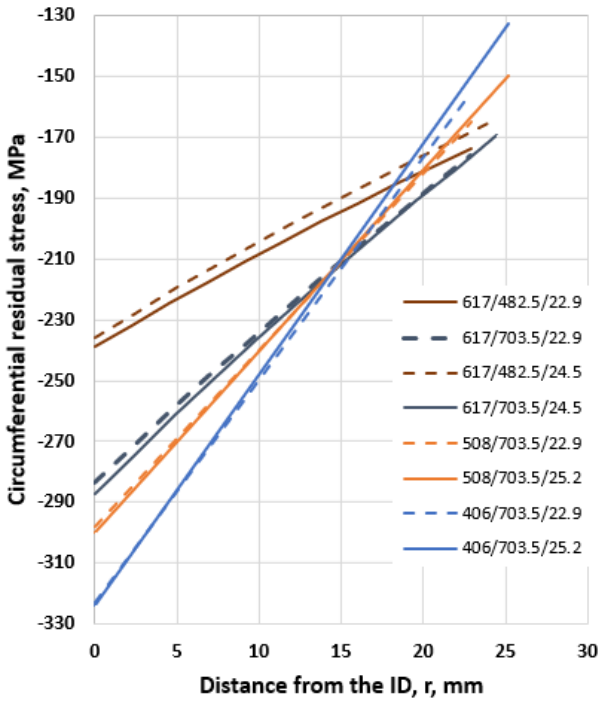


Fig. 2.10 –Residual stress pattern in the liner after autofrettage for cylinders of different ODs and yield strength.

2.2.5 Stresses in the Critical Regions of the Liner During Service

Hoop stress on the ID of the main body of the liner decreases with increase in its thickness from 22.9 to 25.2 mm as seen in Fig. 2.13 for a yield strength of 482.5 MPa. The circumferential stress at this location further decreases with increases in yield strength due to higher residual stresses.

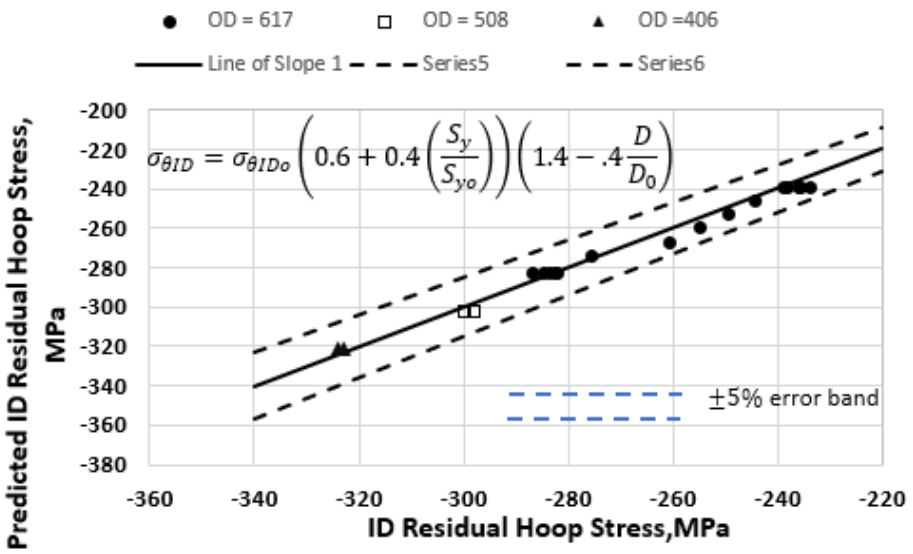


Fig. 2.11 - Predicted residual circumferential stress at the ID from equation (2.4) and that calculated from the finite element analyses under similar conditions.

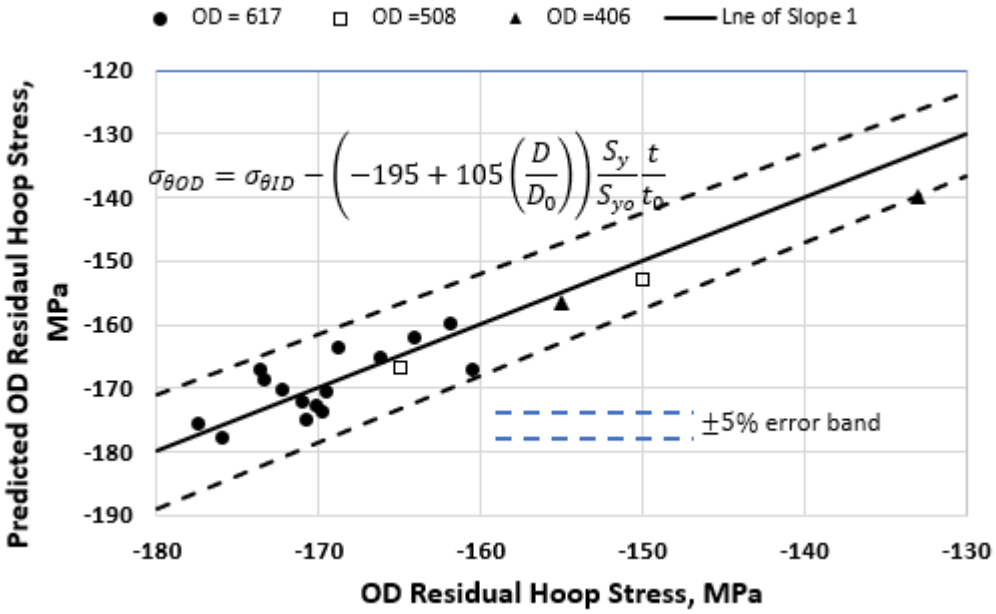


Fig. 2.12 - Predicted residual circumferential stress at the OD from equation (2.5) and that calculated from the finite element analyses under similar conditions.

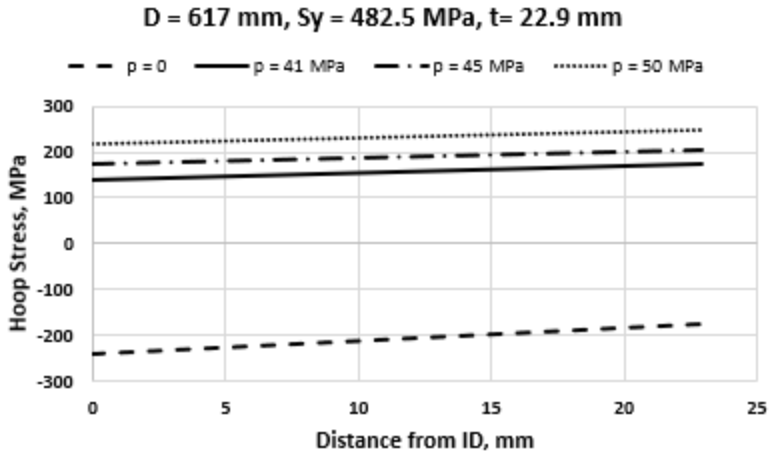


Fig. 2.13 – Circumferential stress distribution between ID and OD of a liner with a thickness of 22.9 mm, and a material yield strength of 482.5 MPa at various operating pressures in the mid-section of the cylinder.

In Fig. 2.14, the circumferential stress distributions from the ID to OD in the mid-region of the cylinder are plotted for liner wall thicknesses of 22.9 and 25.2 mm for an MOP of 50 MPa. As expected, the stresses are lower for the liner with a thicker wall. The effect of OD on the magnitude of the circumferential stress in this region was also explored by considering results from liners with ODs of 508 mm and 406 mm, in addition for the 617 mm OD case. These results are also shown in Fig. 2.14. It is evident that reducing the OD is a much more effective means of reducing the stress in this region than increasing thickness. However, it must be kept in mind that OD has a major influence on the capacity of the cylinder, therefore, reducing it will significantly affect the capacity of the cylinder.

It is noted that the maximum hoop stress on the ID of the liner occurs in the transition region between the straight section and the beginning of the dome. There is a 9 mm region between the end of the wrap and start of the dome that is not reinforced by the wrap and is considered critical. The hoop stress distribution in this region was carefully examined as discussed below.

At a yield strength of 703.5 MPa, the ID and OD circumferential stresses are plotted as a function of axial distance from the wrap/cylinder interface for various wall thicknesses in Figs. 2.15 and 2.16, respectively. The maximum hoop stress at the ID occurs at approximately 1.76 mm from the wrap/liner interface and is shown in Fig. 2.15. The corresponding hoop stress at the OD for that location can be found in Fig. 2.16. The distribution of the circumferential stress from ID to OD at an axial distance of 1.76 mm from the wrap/liner interface is plotted in Fig. 2.17 for two liner thicknesses. It is observed from Figs. 2.15 and 2.16 that when the stress at the ID is high, the corresponding stress on the OD is low and when the stress on the OD is high, the corresponding stress at the ID is low. The effect of this trend on the cyclic life of the cylinder will be discussed later.

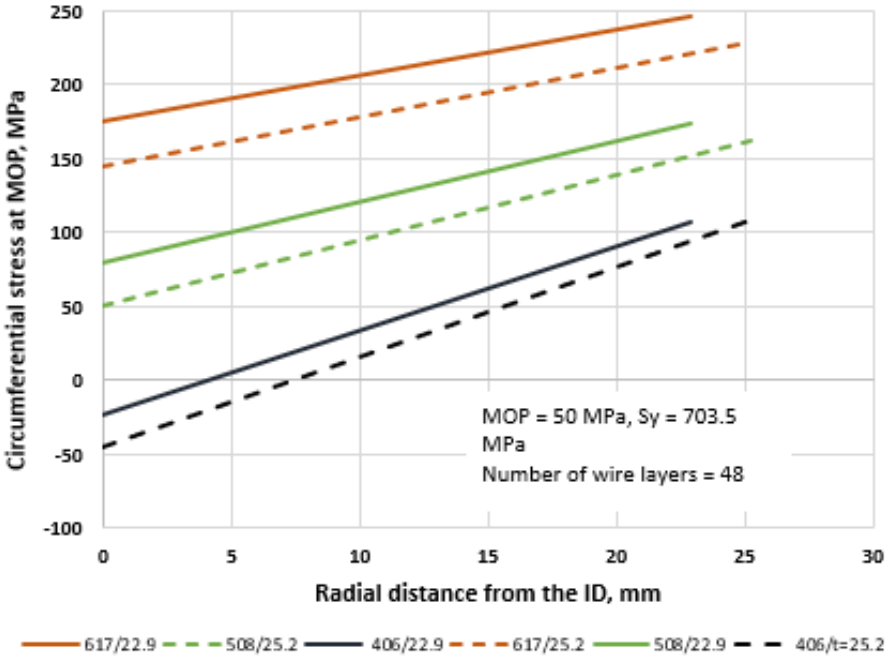


Fig. 2.14 – Circumferential stress distribution for an MOP of 50 MPa in the transition region from the wrap to the dome at an axial distance of 1.76 mm from the liner/wrap interface. The liners analyzed were of three different ODs and two thicknesses, and S_y=703.5 MPa.

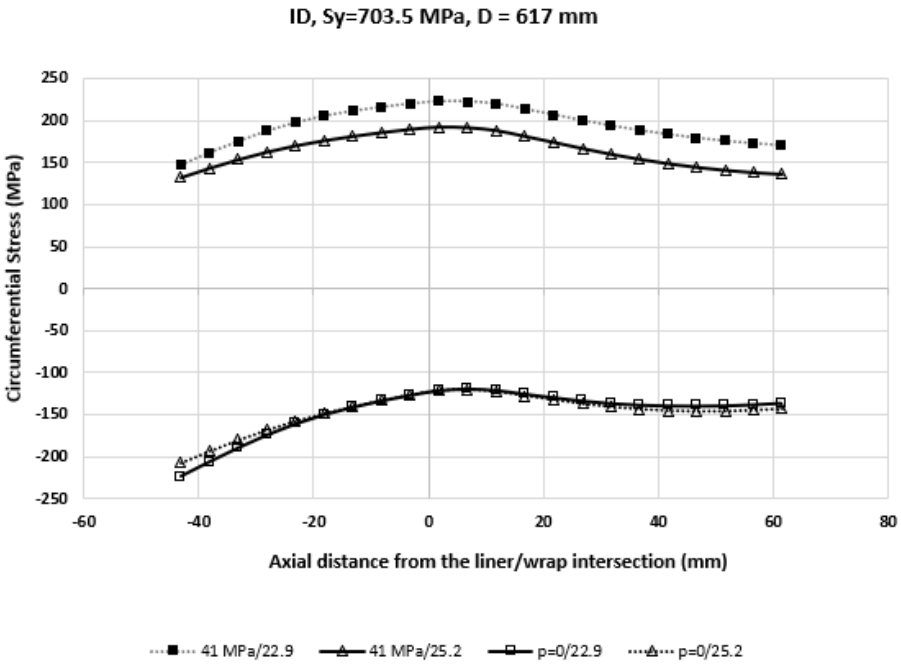


Fig. 2.15 – Hoop stress distribution at the ID at a pressure of 41 MPa and at ambient pressure along the axial direction as a function of axial distance from the liner/wrap intersection for two wall thicknesses. The maximum stress occurs at a point approximately 1.76 mm from the wire wrap/cylinder intersection towards the dome.

Axial stress on the ID were generally low and is not a cause for concern. However, there were regions of the liner where the axial stress on the OD was high as seen in Figs. 2.18 and 2.19. In these regions, the axial stress on the ID is very low. Thus, any cracks present on the ID surface only experience these low stresses. Table 2.4 summarizes the magnitude of stresses at various maximum operating pressures for cylinders with liners of thicknesses equal to 22.9 mm and 25.2 mm, for ODs of 617, 508 mm, and 406mm.

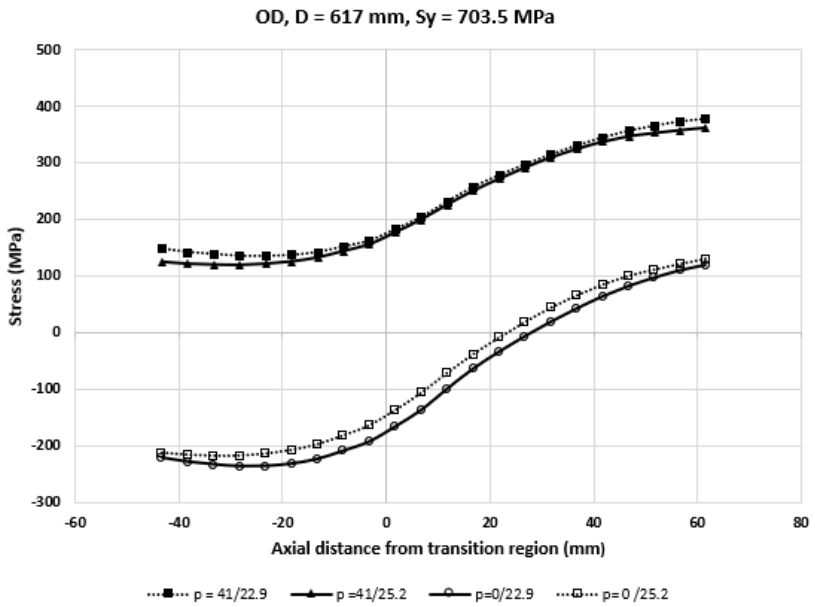


Fig. 2.16 – Maximum hoop stress as function of wall thickness on the OD at a pressure of 41 MPa and at ambient pressure for a yield strength of 703.5 MPa.

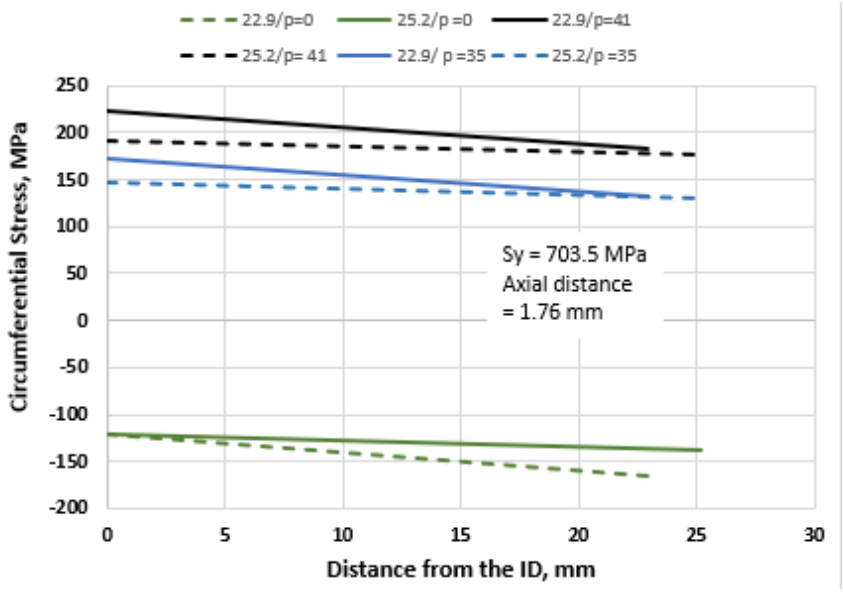


Fig. 2.17 – Circumferential stress distributions from ID to OD at a pressure of 0 and 41 MPa for a 617 mm OD liner at a point 1.76 mm from the wrap/liner interface.

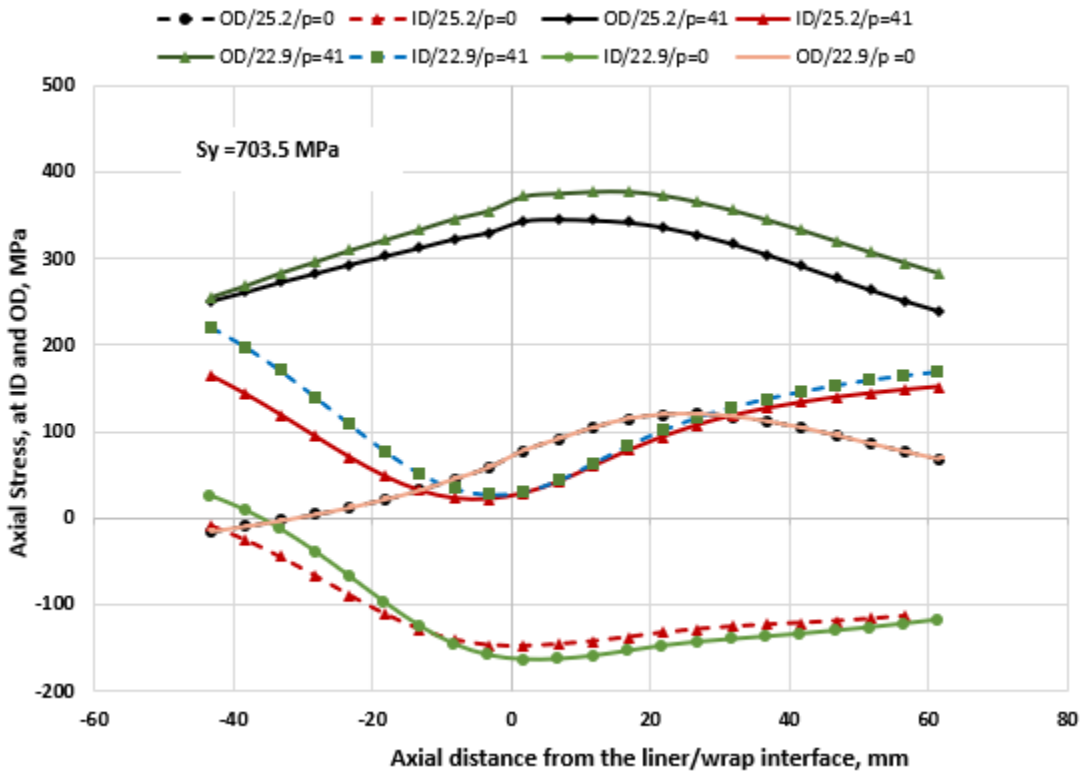


Fig. 2.18- Axial stress at ID and OD of a cylinder subjected to an internal pressure of 0 and 41 MPa for liner thicknesses of 22.9 and 25.2 mm for liners with a yield strength of 703.5 MPa.

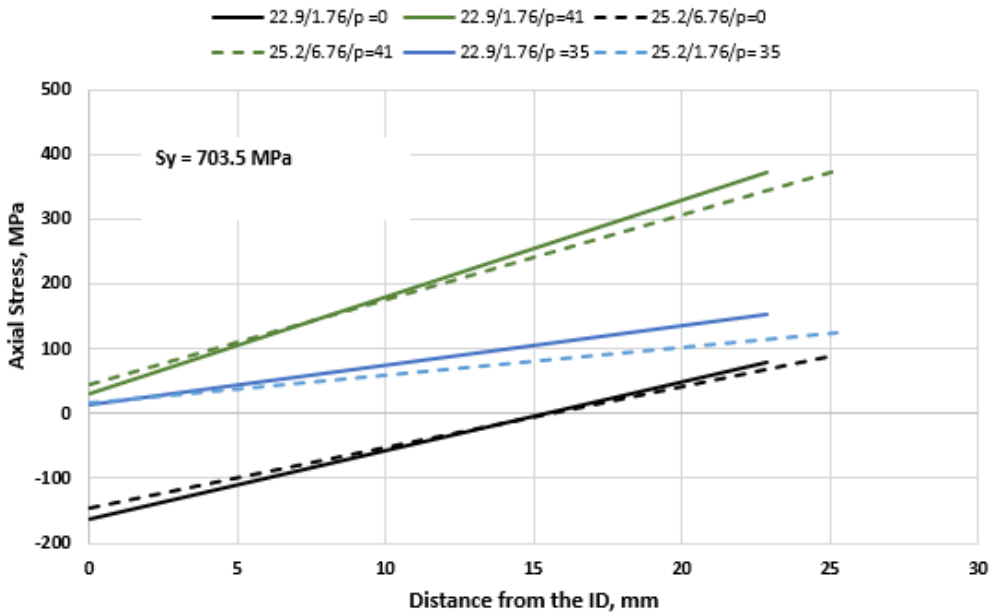


Fig. 2.19 – Axial stress distribution from ID to OD at two locations along the unwrapped portion of the cylinder, at an axial distance of 1.76 and 6.76 mm. The liner OD was 617 mm.

Table 2.4 – Summary of stresses in the critical region in the pressurized cylinder

OD mm	t mm	MOP MPa	Circumferential Stress (mid-section), MPa		Circumferential Stress (transition region), MPa[1]		Axial Stress (transition region), MPa[1]	
			ID	OD	ID	OD	ID	OD
617	22.9	45	129	204	256	216	49.6	401.7
617	25.2	45	101	190	222.3	207.7	46.38	369.6
617	22.9	41	92.3	170.2	222.9	182.3	30.9	372.5
617	25.2	41	66.8	158.8	191.9	177	30.9	343.6
617	22.9	37.5	60.1	141	193.5	155	15.5	347.4
617	25.2	37.5	36.9	131.5	165.1	150.1	14.2	320.8
617	22.9	35	37.2	119.6	172.3	131.2	4.8	329.5
617	25.2	35	15.7	112	146.1	130.9	2.4	304.7
617	22.9	0	-284	-170	-120.4	-166.97	-147.4	78.5
617	25.2	0	-283	-161	-120.4	-137.4	-163.7	78.2
508	22.9	45	41	140	171.8	154.8	-11.6	350
508	25.2	45	15	131	141.4	154.8	-9.75	325
508	22.9	50	79	174	206.2	189.3	-6.8	379
508	25.2	50	50	162	172.7	185.9	-7	351
508	22.9	0	-298	-165	-138.8	-156	-181	85.3
508	25.2	0	-300	-150	-141	-123	-163.6	91
406	22.9	45	-53	81	80	102.3	-7.5	308.6
406	25.2	45	-73	84	55.7	116.8	-64	289.8
406	22.9	50	-23	107	107.5	129.4	-61.2	331.6
406	25.2	50	-45	108	80.7	141	-51.3	310
406	22.9	0	-323	-155	-169	-139.2	-205.3	104.4
406	25.2	0	-324	-133	-170.5	-98.72	-181.0	109.8

[1]These stresses are for a location 1.76 mm in the axial direction from the intersection of the wrap and liner towards the dome of the liner where the stresses were the highest.

In Fig. 2.20, the effect of OD on the magnitude of the circumferential stress in the transition region was explored for two liner wall thicknesses and two wrap thicknesses. The wrap thicknesses analyzed were 17.06 (48 layers of wire) and 14.93 mm (42 layers of wire). As expected, the stress decreases significantly with decreasing OD and moderately with increasing wrap and liner thicknesses. Between wrap thickness and liner thickness, the decrease in stress is more significant for increasing wrap thickness. Thus, it is better to have a wrap thickness of 17.06 mm than 14.93 mm.

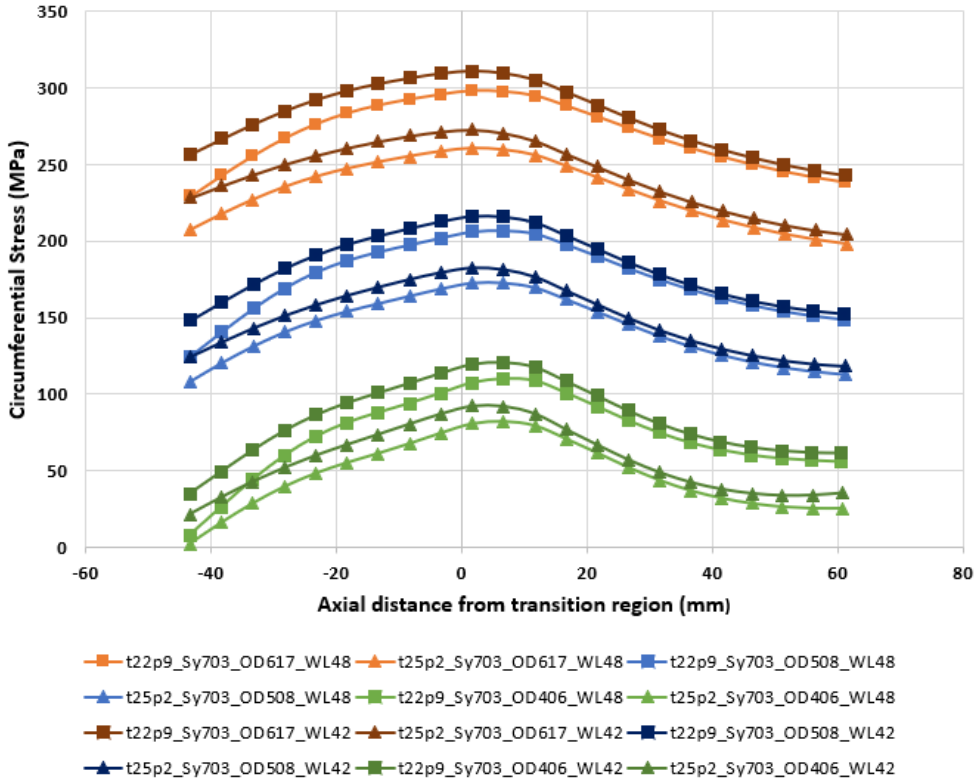


Fig. 2.20 – The effect of OD, liner wall thickness, and the wrap thickness on the circumferential stresses in the transition region of the liner.

2.3 Objective 3: Fracture Mechanics Analyses

The results of the Fracture Mechanics analyses to estimate the design life of the cylinder are described. These analyses results were used to choose the liner and other cylinder parameters for the cylinder prototype. First, the crack growth material properties used in determining the design life are summarized.

2.3.1 Environment Assisted Fatigue Crack Growth During Service Loading

Environment assisted fatigue crack growth analysis was performed for a variety of conditions that included MOP values of 41 and 35 MPa (410 and 350 bar) for cylinders containing liners with 617 mm OD. In these analyses, the postulated cracks were in the radial-axial plane in the transition region where the circumferential stresses were the highest. The initial crack shapes assumed were semi-elliptical; they were 1 mm deep and 3 and 5 mm long on the surface. The material model chosen for the environment assisted crack growth analysis is first described, followed by the results of the design life calculations. The results for a cylinder with a liner of diameter 508 mm are discussed that will permit raising the MOP to 50 MPa (500 bar).

The ASME master fatigue crack growth rate curve embedded in ASME Code Case 2938 was developed by San Marchi et al.[4] of Sandia National Laboratory. It was used in the fracture mechanics analyses presented here as it applies to ferritic steels used in

the construction of pressure vessels experiencing high pressure hydrogen environment during service. The salient features of this model are as follows:

- Fatigue crack growth rates in A372 Grade J (or E) Class 70 steels with ultimate strength less than 950 MPa (137 ksi) are relatively insensitive to heats of steel and the yield strength. Also, the data exhibit a knee in the da/dN versus ΔK trend at a value of ΔK equal to ΔKc, that depends on the load ratio, R. The slope in the Paris equation changes from 6.5 to 3.66 at ΔKc as seen in Fig. 2.21 for various load ratios.

- The FCGR behavior increases significantly with load ratio, $R = K_{min}/K_{max}$; thus, the effects of R must be accounted in calculating crack growth rates in gaseous hydrogen. A relatively simple relationship, equation (2.6), can represent the fatigue crack growth rate (FCGR) behavior of this class of steels for load ratio, R, $0.1 \leq R \leq 0.8$ These equations are used for hydrogen pressures of up to 102 MPa (15,000 psi) and are as follows:

$$\frac{da}{dN} = C\frac{1+C_H R}{1-R}(\Delta K)^m \tag{2.6}$$

$$\Delta K_c = 8.475 + 4.062R - 1.69R^2 \tag{2.7}$$

where, the values of C, C_H, and m are listed in Table 2.5. Note that C and m values characterize the dependence of crack growth rates on ΔK in a linear log-log Paris-type relationship, and the value of C_H is dependent on R.

The FCGR data from San Marchi et al.[4] is limited to $R > 0$; but autofrettaged vessels are subjected to loading that consists of compressive stresses at the minimum pressure leading to an $R < 0$. The ASTM Standard E647 [5] recommends that for $R < 0$, ΔK equal K_{max}. The ASME Code Case 2938 also recommends that for R < 0, ΔK equal K_{max} or essentially, $R = 0$.

Table 2.5 – Constants in equation (2.6) for describing the environment assisted fatigue crack growth behavior in SA372 Grade E/J Class 70 steels in high pressure hydrogen (p = 102 MPa)

	$\Delta K < \Delta K_c$		$\Delta K \geq \Delta K_c$	
	da/dN in in/cycle	*da/dN* in m/cycle	*da/dN* in in/cycle	*da/dN* in m/cycle
C	2.54×10^{-12}	3.5×10^{-14}	8.34×10^{-10}	1.5×10^{-11}
m	6.5		3.66	
C_H	0.43		2.0	

A comparison of FCGR behavior between positive and negative load ratios in gaseous hydrogen at a pressure of 10 MPa was investigated by Saxena, Nibur and Prakash [6]. They concluded that the FCGR behavior for R of -0.5 and -1.0 were comparable to the trends exhibited at $R = 0.1$ tested under identical hydrogen pressures

and using specimens that were suited for both tension-tension and tension-compression loading. This evidence is used as a justification for using R = 0 in equation (7) when the load ratios are negative, as also specified in ASME Code Case 2938.

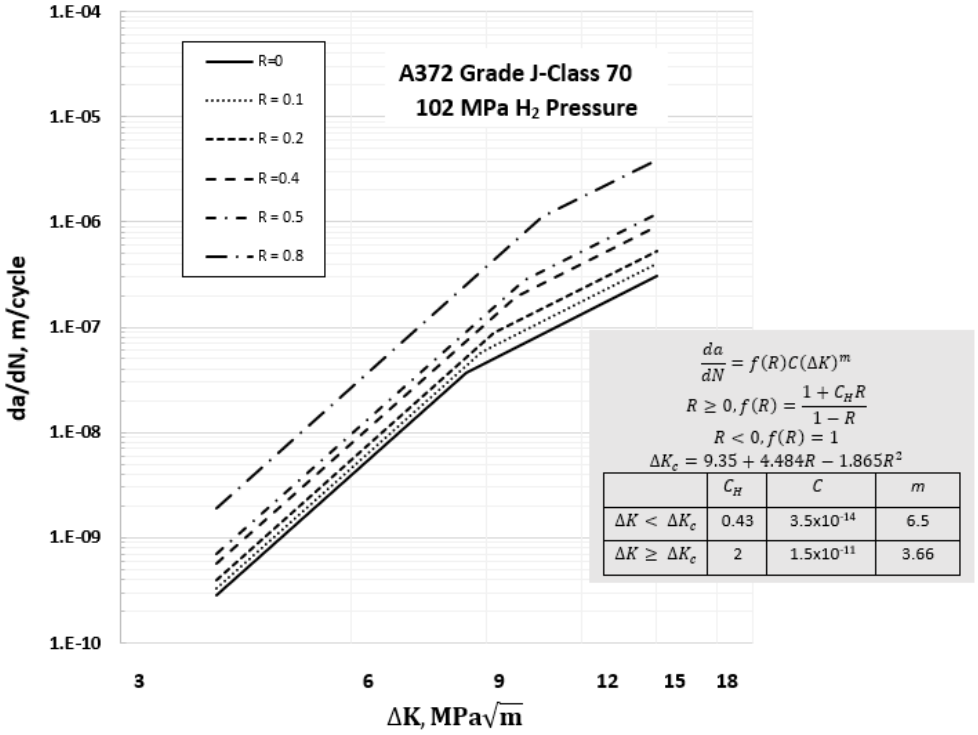


Fig. 2.21- Schematic representations of the FCGR behavior from the ASME Section VIII-3 master curve for A372 Grade J/E Class 70 steels as a function of load ratio, R, in high pressure gaseous hydrogen for pressures of 100 MPa.

2.3.2 Design Life Estimation

In this section, the design lives of the pressure vessels were estimated assuming flaw sizes in liners that could escape detection during the ultrasonic and magnetic particle inspections of liners after manufacturing. The geometry of the flaw that is life limiting is shown in Fig. 2.22. As presented in the previous section, the highest circumferential stresses on the ID of the liner that is exposed to high pressure hydrogen is at a point located at an axial distance of 1.76 mm from the end of the wrap toward the dome. This is a region not covered by the wrap. The magnitude of the stresses in the ID and OD at this location were listed in Table 2.3 for this location. There are other critical areas in the liner of the wrapped cylinder near the knuckle region where the stresses are high and will be explored later in the report. The stresses at this location are used only for preliminary design and the other locations will be analyzed for estimating the design life of the vessel as part of certification and preparing the manufacturer's design report (MDR).

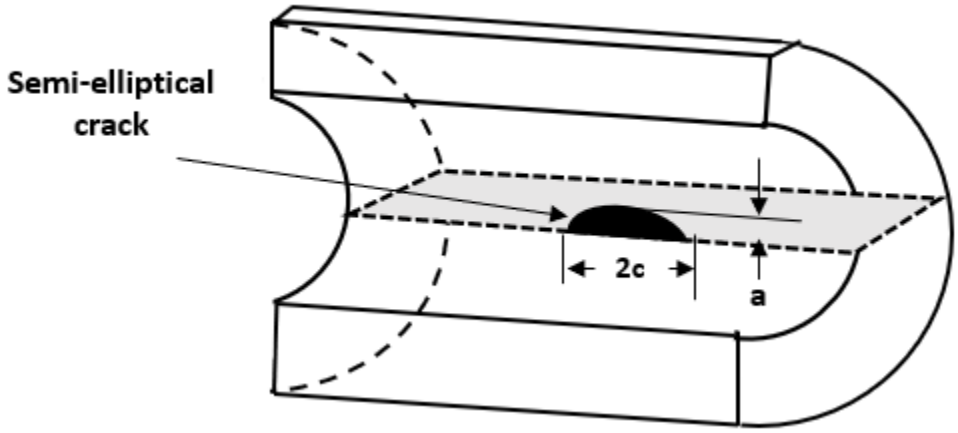


Fig. 2.22 – The geometry of the flaw that determines the design life of the vessel.

To estimate the design life, the magnitude of the stress intensity parameter, K, for this geometry must be estimated. Expressions derived by Newman and Raju [7] are adopted for this purpose. These expressions are summarized below:

$$K = (S_t + p + H(\phi)S_b)\sqrt{\frac{\pi a}{Q}}F \tag{2.8}$$

Where, p = gas pressure (MOP) and the other terms are as follows:

$$F = F_1 f_c g(\phi) f(\phi) \tag{2.9}$$

Where, ϕ is the angle as shown in Fig. 2.23. The various other terms in equation (2.9) are given as:

$$Q = 1 + 1.464(a/c)^{1.65} \text{ for a/c < 1}$$

$$F_1 = [M_1 + M_2(a/t)^2 + M_3(a/t)^4]$$

$$M_1 = 1.13 - 0.09\frac{a}{c}$$

$$M_2 = -0.54 + \frac{0.89}{0.2 + \frac{a}{c}}$$

$$M_3 = 0.5 - \frac{1}{0.65 + \frac{a}{c}} + 14\left(1 - \frac{a}{c}\right)^{24}$$

$$g(\phi) = 1 + \left[0.1 + 0.35\left(\frac{a}{t}\right)^2\right](1 - sin\phi)^2$$

$$f(\phi) = \left[sin^2\phi + \left(\frac{a}{c}\right)^2 cos^2\phi\right]^{1/4}$$

$$f_c = \left[\frac{r_o^2 + r_i^2}{r_o^2 - r_i^2} + 1 - 0.5\sqrt{\frac{a}{t}}\right]\frac{t}{r_i}$$

$$g(\phi = 0) = 1 + \left[0.1 + 0.35\left(\frac{a}{t}\right)^2\right]$$

$$g(\phi = \pi/2) = 1$$
$$f(\phi = 0) = \sqrt{\frac{a}{c}}$$
$$f(\phi = \pi/2) = 1$$
$$H(\phi = 0) = H_1$$
$$H(\pi/2) = H_2$$
$$H_1 = 1 - 0.34\frac{a}{t} - 0.11(a/c)(a/t)$$

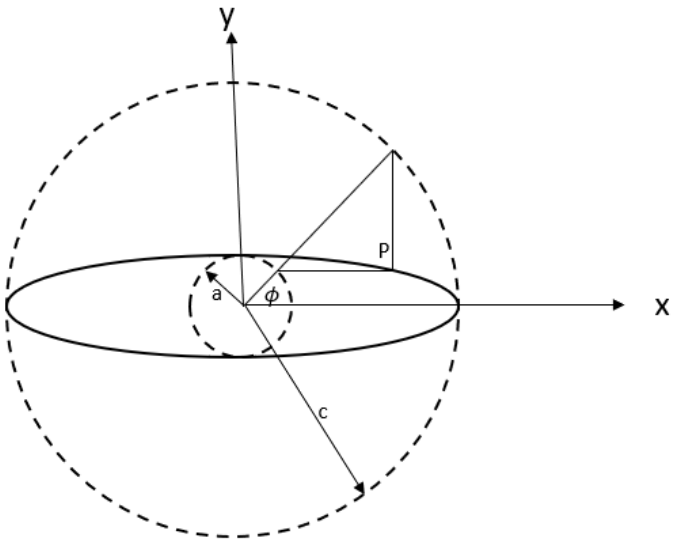


Fig.2. 23- Definition of the angle ϕ at any point P on the boundary of an elliptical crack. The value of $\phi = 0$ on the surface and 90° at the deepest point.

$$H_2 = 1 + G_1(a/t) + G_2(a/t)^2;\ G_1 = -1.22 - 0.12(a/c)$$
$$G_2 = 0.55 - 1.05\left(\frac{a}{c}\right)^{0.75} + 0.47(a/c)^{1.5}$$
$$S_t = \frac{\sigma_{\theta ID} + \sigma_{\theta OD}}{2};\ S_b = \frac{\sigma_{\theta ID} - \sigma_{\theta OD}}{2}$$

The above equations are embedded in a computer code for estimating design lives. Comparisons made between the results from this code and those with NASGRO [8], a commercial Fracture Mechanics analysis code used widely in industry, showed that the WireTough code based on the above equations always resulted in conservative lives. The code was implemented for the following conditions:

- The liner size consisted of an OD = 617 mm and thicknesses of 22.9 and 25.2 mm wrapped with 48 layers of wire. In other words, the wrap thickness is 17.06 mm.
- Hydrogen pressure ranges varied between 45 to 4.5 MPa, 41 and 4.1 MPa, 37.5 to 3.75 MPa, 35 to 3.5 MPa, to determine how the design lives vary with MOP under conditions of full cycling.
- The initial flaw sizes were chosen as 1 mm deep and 0.5 mm deep. The 1 mm deep flaw is currently an industry standard based on results of ultrasonic testing,

but the industry is moving rapidly to a capability of being able to detect flaws that are 0.5 mm deep on the ID of the liner. Thus, in this analyses both flaw depths are considered.

- The initial flaw depth, a, to surface flaw length, 2c, were chosen as 0.2 and 0.33 to assess the differences in the design life from being able to detect smaller flaws, not just in the depth direction but also along the axial direction on the surface.

The results of the parametric analyses are reported in Table 2.6 and plotted in Fig. 2.24. The following criteria developed from the recommendations of ASME Section VIII-Division 3 design code KD-10 for pressure vessels designed for service in high pressure hydrogen environment were followed to determine design life.

- The crack growth calculation was truncated when the maximum value of K, K_{max}, reaches a value of 40 MPa$(m)^{0.5}$ corresponding to the threshold stress intensity level for crack growth under a sustained load. This is considerably lower than the estimated fracture toughness, K_{Ic} = 80 MP$(m)^{.5}$
- The design life was equal to half of the number of cycles needed to grow the crack from its initial flaw depth to the critical flaw depth.
- In all cases, the K along the flaw length was significantly less than the K along the depth direction, so a leak-before-break condition is assured.

At an MOP of 45 MPa (450 bar), to achieve 4000 cycles of design life requires a liner thickness of 25.2 mm and the ability to detect flaws that are 0.5 mm deep. This is currently not within the capability of the currently used ultrasonic inspection techniques in industry. Therefore, 45 MPa cannot be recommended as the MOP at this time.

At an MOP of 41 MPa (410 bar), a cylinder with a liner of wall thickness of 22.9 mm will not yield an acceptable life but a cylinder with a liner that is 25.2 mm thick does for an initial flaw depth of 1 mm. At an MOP of 37.5 MPa (375 bar), the design criteria are met with plenty of margin, Therefore, an MOP of 37.5 MPa (375 bar) is appropriate for the cylinder containing the selected liner with a minimum wall thickness of 22.9 mm. Note that if the cylinder is designed to cycle between MOP and 10% of MOP 200 times in a year, the total desired life over 20 years must be 4,000 cycles.

Next, we consider the liner size needed to raise the MOP to 50 MPa (500 bar) and the MAWP to greater than 60 MPa (600 bar). The most effective means of increasing the MOP to 50 MPa (500 bar) and the maximum allowable pressure (MWAP) to greater than 60 MPa (600 bar) is to reduce the OD of the liner to 508 mm. To partially compensate for the decrease in cylinder capacity, the length is increased to 9000 mm. With that change, the cylinder capacity becomes approximately 1400 L compared to the previous value of 1715 L. Because of the higher H_2 density at 500 bar compared to 350 bar, the mass of hydrogen stored in this cylinder is 42 KG that is higher than 40.7 KG in the larger cylinder at 350 bar. The design lives under these circumstances are explored below to show that this is a viable design.

Table 2.6 – Design lives resulting from the parametric analyses that varied MOP, wall thickness, initial flaw size, and initial flaw shape. The liner OD in all these cases was 617 mm with a wrap thickness of 17.06mm.

MOP, MPa	Liner Thickness, mm	Initial Flaw depth, mm	Design Life, Cycles	
			a/ac = 0.2	a/2c = 0.33
45	22.9	1.0	1733	2390
		0.5	3459	4583
	25.2	1.0	2524	3498
		0.5	5069	6759
41	22.9	1.0	2794	3850
		0.5	5572	7207
	25.2	1.0	4170	5749
		0.5	8799	11454
37.5	22.9	1.0	4634	6383
		0.5	9306	12160
	25.2	1.0	7025	9783
		0.5	14293	20096
35	22.9	1.0	6868	9467
		0.5	13850	19189
	25.2	1.0	10675	14685
		0.5	21343	35064

From Table 2.1, it can be noted that a 22.9 mm thick wall liner with an OD of 508 mm will allow an MAWP of up to 62 MPa (620 bar) and a liner with a wall thickness of 25.2 mm will allow an MAWP of up to 68.7 MPa (687 bar). Thus, liners with an OD of 508 mm of both thicknesses will be sufficient to meet the MAWP goal. We then need to assess if these liner dimensions are sufficient to meet the MOP goal of 50 MPa.

The calculated design lives of cylinders with a liner OD of 508 mm and wall thicknesses of 22.9 and 25.2 mm with a wrap thickness of 17.06 mm were determined for MOP values of 45 and 50 MPa and are listed in Table 2.7 and plotted in Fig.2.25. At an MOP of 45 MPa (450 bar), both thicknesses are sufficient to meet the design life goal of 4000 cycles for flaws that are 1 mm deep and 5 mm long. At MOP of 50 MPa, liner thickness of 25.2 mm meets the design life goal of 4,000 cycles. It is thus necessary to specify a minimum liner wall thickness of 25.2 mm for a liner with an OD of 508 mm to comfortably meet the 50 MPa (or 500 bar) goal for the MOP.

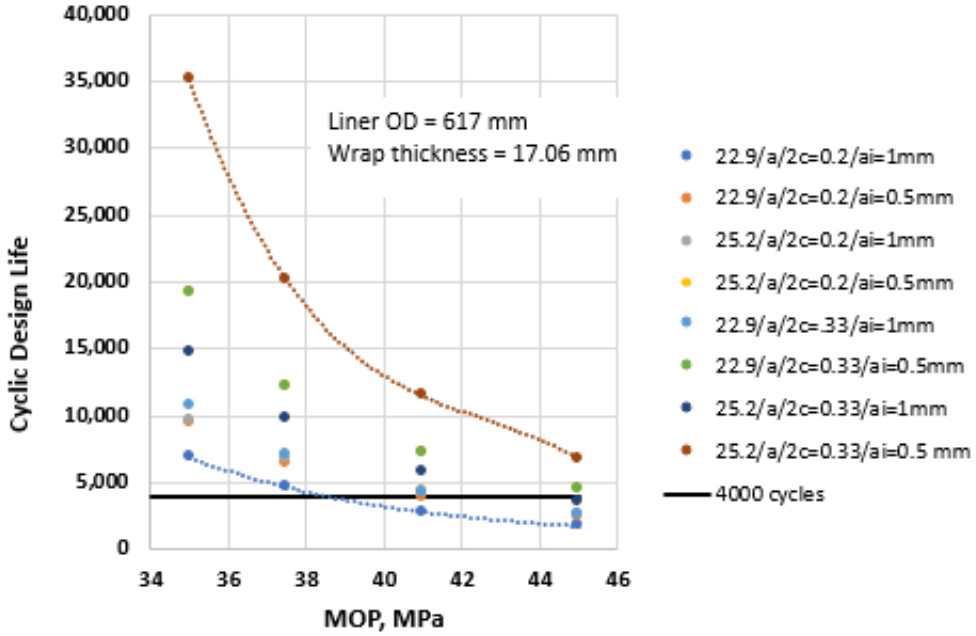


Fig. 2.24 – Design life of cylinders with 617 mm OD liners for full pressure fatigue cycles as a function of MOP, initial flaw shapes and initial flaw size.

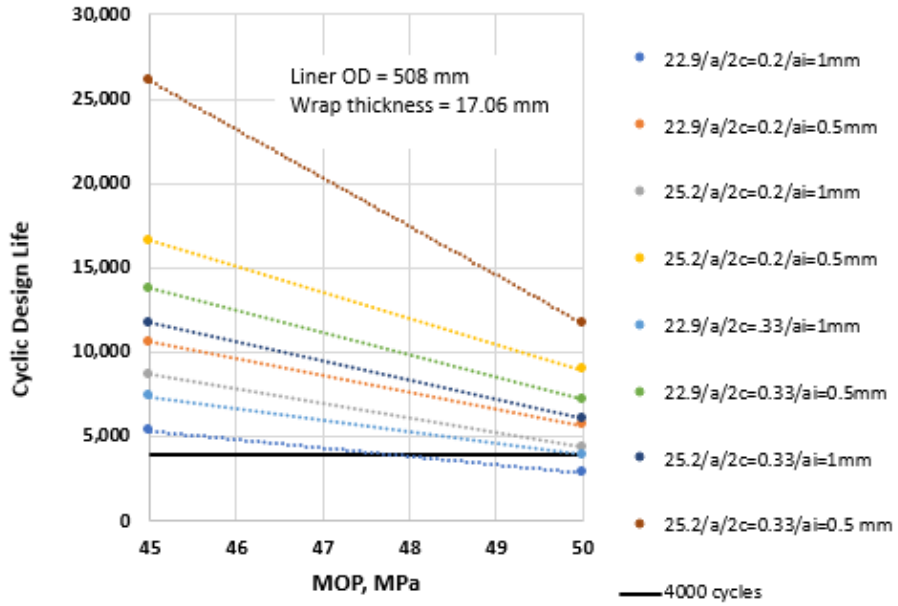


Fig. 2.25 – Design lives of cylinders with 508 mm OD liners and a wrap thickness of 17.08 mm (48 wire layers) as a function pf MOP for wall thicknesses of 22.9 and 25.2 mm, and initial flaw depth of 0.5 mm and 1 mm and a/2c values of 0.2 and 0.33.

Table 2.7 – Design life of cylinders with liner diameters of 508 mm and a wrap thickness of 17.06 mm at MOPs of 45 and 50 MPa.

Wall thickness, t. mm	Initial crack depth, a_i, mm	MOP, MPa	Design Life, Cycles	
			a/2c = 0.2	a/2c = 0.33
22.9	1.0	45	5341	7357
	0.5		10614	13775
25.2	1.0		8662	11731
	0.5		16618	26079
22.9	1.0	50	2862	3945
	0.5		5687	7214
25.2	1.0		4375	6094
	0.5		8937	11731

2.4 Summary of Design Analysis and Fracture Mechanics Analyses Results

Stresses were estimated in the critical region of the vessel subject to service loading consisting of pressurization and depressurization cycles. The circumferential stresses near the junction between the liner and the wrap were the highest and were used to estimate the environment assisted fatigue crack growth life of the vessels. The analyses lead to the following conclusions.

- Autofrettage, used during fabrication of Type II-S pressure vessels to enhance the fatigue life of vessels, locks high compressive residual stresses in the vessel's liner wall and reduces the maximum and the mean stress during service loading.
- To achieve a consistent residual stress profile across the wall, autofrettage pressure must be selected such that there is plastic deformation on the outside surface of the liner. The pressure depends on liner wall thickness and the wrap thickness, the OD of the liner, and the liner yield strength.
- From the finite element analyses, a model was developed to represent the effects of these variables on the autofrettage pressure in the form of a simple, closed form equation.
- The resulting residual stress profiles from autofrettage obtained from finite element analyses were used to develop equations for estimating the residual stress distribution as a function radial distance along the wall thickness, liner OD, and the yield strength of the material.
- To achieve a lifetime of 20 years in pressure vessels that will experience an average of 200 cycles of pressurization/depressurization cycles per year, it is necessary to limit the maximum operating pressure to 41 MPa (410 bar) for vessels with an OD of 617 mm and liner wall thickness of 25.2 mm and to 37.5 (375 bar) for vessels with liner wall thickness of 22.9 mm.

- It will be shown later in the report that a flaw on the ID in the knuckle region between the dome and the nozzle gives the shortest life and limits the MOP to 350 bar. This additional analysis was performed as part of design certification and preparation of the Manufacturer's Design Report (MDR) described in Section 2.6 of this report.
- To achieve an MOP of 50 MPa, it is necessary to decrease the liner OD to 508 mm and raise the minimum wall thickness of the liner to 25.2 mm.
- To achieve an MOP of 700 bar, the liner OD of the liner should be decreased to 406 mm and a minimum wall thickness be increased to 31.75 mm.

2.5 Objective 4: Cylinder Manufacturing

Handling of and wire winding of large cylinders 650 mm OD, up to 7,600 mm long, and weighing almost 4 metric tons is challenging. The manufacturing steps consist of assuring the liner meets all the specifications of ASME Section VIII-Division 3 for liners and must be sourced from an ASME approved manufacturing facility. Subsequently, the wire winding in WireTough's facility must be performed in accordance with ASME approved, WireTough prepared quality assurance (QA) manual and the Manufacturer's Design Report (MDR) prepared by a third party ASME approved contractor who in this case was Structural Integrity Associates with head offices in California. The details of manufacturing processes and cost analysis are described next.

Among the three possible cylinder sizes, the one with a MOP of 350 bar and MAWP of 500 bar was chosen for developing and demonstrating the manufacturing technologies and building a prototype of the designed vessel. This design was the largest in terms of OD size and capacity of the cylinder and the cylinder had the lowest weight of the cylinder per KG of hydrogen stored.

2.5.1 Liner Procurement

Tenaris USA was chosen to manufacture a custom liner with the following specifications.

- The liner is made from SA372 Grade E/J Class 70 seamless steel pipe with swaged ends. No welding of any kind is permitted.
- Quenched from a solution anneal temperature of 940 ^{0}C and tempered at 625 ^{0}C.
- The OD of the liner = 610 $\pm 1\%$ mm, length = 7600 $\pm$ 50 mm, and minimum wall thickness = 22.9 mm.
- The nominal weight of the liner = 2960 $\pm$ 10% KG

An order for two identical liners was placed with Tenaris in December 2021 and the finished custom liners were delivered to WireTough in April 2023. A materials test report including results from non-destructive-evaluation (NDE) testing was provided by Tenaris. A picture of the finished liners is shown in Fig. 2.26. Schematic pictures of the dome regions of the liner are shown in Fig. 2.27 where wall thickness measurements were made to ensure that the liners were acceptable. The thickness in the straight section of the liner

was measured to be between 26.18 to 29.33 mm that significantly exceeds the minimum required value of 22.9 mm. Because of the metal flow during forming of the domes, the thickness in the transition region from the dome to the nozzle is even higher between 38.63 to 43.3 mm. The implications of these measurements will be discussed further in Section 2.6.

The chemical composition of the steel of heat number of 920398 from which the liners were made, and the standard composition limits are shown in Table 2.7. All compositions were within the specified range.

Table 2.8 – Chemistry of the liner steel and the acceptable composition range (wt %).

Heat No.	C	Mn	Si	Cr	Mo	P	S	S + P
Standard	0.25 -0.4	0.4 – 1.0	0.1 – 0.45	0.8 – 1.2	0.15 -0.35	0.025 max	0.025 max	0.03 max
920398	0.28	0.77	0.21	0.99	0.21	.013	0.002	0.015

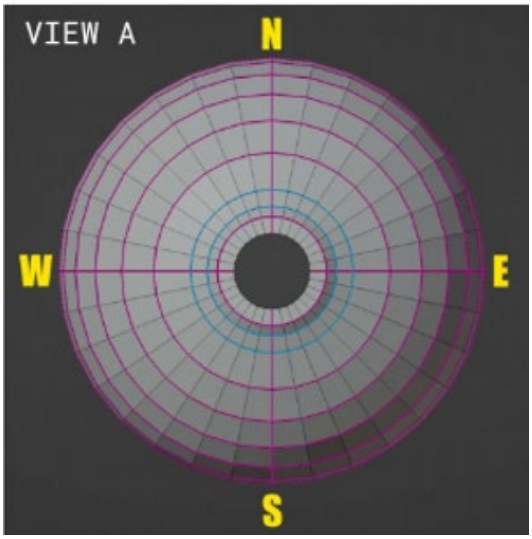


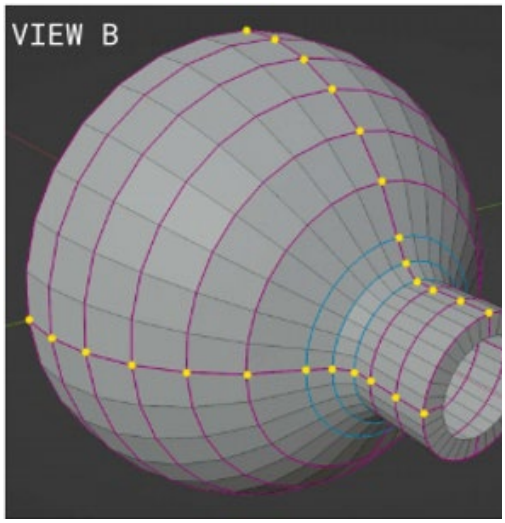
Fig. 2.26 – Picture of the two Tenaris seamless liners being delivered to WireTough's facility.

The tensile properties of the steel are shown in Table 2.9 and compared to the specified ranges. The dimensions of both liners as well as the mechanical properties were all within specifications.

Table 2.9 – Tensile test results of the liner material.

Material Heat No.	Test Temp, ^{0}C	YS, MPa	UTS, MPa	% Elongation (50 mm gage length	CVN (Joules) At -40 ^{0}C	Brinell Hardness
Standard range	20 – 25 ^{0}C	>485	825 – 950	>16	50 (Avg)	250 - 300
92038	>20 ^{0}C	709	844	19	113 (102 – 121)	262





Location	Wall thickness, mm			
	North	South	East	West
1	27.0	25.68	27.6	27.6
2	29.33	26.36	27.22	26.87
3	33.5	26.62	29.89	26.87
4	39.5	31.65	35.10	34.12
5	42.2	38.63	40.00	34.12
6	43.23	41.25	41.68	39.37
7	38.89	41.37	41.48	40.61
8	39.19	38.5	39.01	38.63
9	26.18	29.59	39.01	36.78
10	29.77	26.06	26.00	26.36
11	29.84	28.64	29.72	29.72
12	29.72	29.77	29.72	29.72

Fig. 2.27 (Top) – Schematics of the dome regions of the liners showing points where thickness measurements were made, (bottom) results from those measurements in multiple locations.

2.5.2 Wire Wrapping of the Cylinder

The wire winding machine used was a McClean Anderson and Magnum Venus Products (MVP) capable of winding liners that are up to 10,000 mm long. The liners were wound with 48 layers of wire following the procedure outlined in WireTough's ASME approved QA manual. A picture of the wound liner with a UV polymer coating is shown In Fig. 2.28. The wire used for winding was SA905 plain carbon steel with a diameter of 0.35 mm with a tensile strength greater than 3 GPa. The ASME Code Case limits the yield strength of the wire to 1.8 GPa. In other words, the stress in the wire must not exceed 1.8 GPa during the autofrettage process.



Fig. 2.28 – Picture of a wrapped cylinder coated with a UV resistant coating ready for autofrettage and hydrotesting.

The cylinders were shipped to Authorized Pressure Testing in California for autofrettage and testing. The pressures used for autofrettage, and testing are summarized in Table 2.10 below. The formula for estimating Autofrettage Pressure for 48-layer steel wire was given in equation (2.3). Hydro Test Pressure as per MDR = 1.33 x 51 MPa (design pressure) = 66.9 MPa.

Table 2.11 lists all the construction materials in building the Type II-S wire wrapped vessel. All construction materials are readily available commercially.

Table 2.10 – Calculated autofrettage pressures and hydrotest pressures for the two cylinders built.

WTC Serial Number	Liner ID		D (mm)	Wall thickness, t (mm)	S_y (MPa)	P_A (MPa)
	Manufacturer	**Serial Number**				
WTC_C6107600-001	Tenaris	JC-056677	611.5	27.4	708.7	117.7
WTC_C6107600-002	Tenaris	JC-056675	611.78	1.045	102.8	17.152

Table 2.11 – List of materials of construction for manufacturing Type II-S wire wrapped vessel.

Item	Commercial Source
Liner (SA272 Grade E/J – Class 70 -Pressure Vessel Steel	Tenaris Dalmine SPA, TX CPI McKeesport, PA
SA 905 0.35 mm diameter round Wire	Kiswire, Japan with ware house in Pine Bluff, Arkansas
Epoxy resin from 3M	3 M, Minneapolis, MN
UV Protective skin	Pittsburgh Paint and Glass, Pittsburgh, PA

2.5.3 Estimation of Cylinder Manufacturing Costs

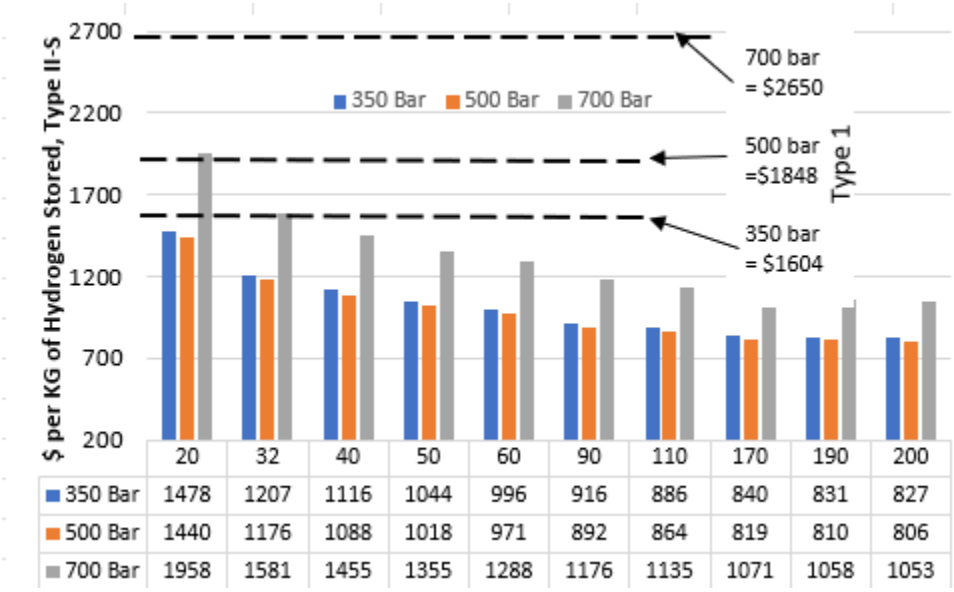
A single assembly line consisting of a winder, material handing system, creels for feeding the wire and the autofrettage and hydrotest facility working in two-shifts can produce up to 200 wire-wound cylinders per year. The fixed annual cost for operating one manufacturing line is estimated as $593,000 and it includes:

- Salaries and incentive bonuses for 33% of a Mechanical Engineer, 33% Product Manager, 33% of a QA Manager, and 3 full-time shop floor technicians.
- Fringe benefits and pay roll taxes for the above personnel,
- Cost of contractors for maintaining certifications, cost of professional engineering services, and securing ASME Authorized Inspector services in support of the manufacturing operation.
- Administration and accounting,
- Marketing related travel,
- Legal and insurance expenses,
- Facility rental for required manufacturing space for one line,
- Costs of repair and equipment replacement,
- Office supplies and communications.

The fixed annual costs detailed above are distributed equally among cylinders as part of annual production volume (APV). For estimating cylinder costs, the APV was varied between 20 to the full capacity of 200 cylinders. Note that the fixed cost does not include initial equipment or set-up costs such as the cost of the winder and its material handling accessories and the autofrettage/hydrotest facility and installation.

The variable costs include the cost of materials as listed in Table 2.10, warranty reserves, cost of working capital, and inbound shipping costs of materials. This cost is assumed to be the same for a given cylinder type (fixed pressure rating and capacity) that scales linearly with APV. In other words, no discounts were included for bulk purchase of construction materials. Figure 2.29 shows how the fixed, variable, and total costs vary with APV for cylinders with pressure ratings of 350, 500 and 700 bar. Since each type of cylinder stores a different amount of hydrogen, a better metric for comparing costs across cylinders with different storage capacity is the cost per KG of hydrogen stored as used in Fig. 2.29. Prices of Type 1 cylinders are also shown in Fig. 2.29 for comparison.

As expected, the cost per cylinder reduces significantly with increasing APV for cylinders of all pressure ratings. There is potential for cost savings of between 48.5% and 60 % to be realized by switching from Type 1 design to a Type II-S design that is bulk produced. The cylinder weight per KG of H_2 stored for the Type 1 and Type II-S cylinders with different pressure ratings is shown in Table 2.12. The weight savings in switching from Type I to Type II-S range from 55% to 62.5%.



	20	32	40	50	60	90	110	170	190	200
350 Bar	1478	1207	1116	1044	996	916	886	840	831	827
500 Bar	1440	1176	1088	1018	971	892	864	819	810	806
700 Bar	1958	1581	1455	1355	1288	1176	1135	1071	1058	1053

Fig. 2.29 – Cost of hydrogen storage in $ per KG of hydrogen stored using Type II-S cylinders as a function of Annual Production Volume in a single production line. The cost is compared to the cost of using Type 1 cylinders.

Table 2.12- Cost and weight savings in switching from Type 1 to Type II-S cylinders for MOP of 350, 500, and 700 bar.

MOP, Bar	Cost/KG of H_2 Stored ($)			Cylinder Weight/KG of H_2 Stored, KG		
	Type II-S	Type 1	% Savings	Type II-S	Type 1	% Savings
350	827	1604	48.5	90.4	200.5	55.0
500	806	1848	56.3	93.5	231.0	59.5
700	1053	2650	60	122.1	325.4	62.5

2.6 Objective 5: Obtain ASME U3 Certification for the H_2 Storage Vessel

The vessel that was designed for 350 bar MOP and 500 bar MAWP (design pressure), and a capacity of 1715 L was taken through the entire ASME certification process. Two cylinders were produced as described in the previous section of this report. Prior to making these cylinders, the design was evaluated by an ASME authorized third-party consultant, Structural Integrity Associates (SIA), and certified.

The structural capacity of the vessel was shown to meet the requirements prescribed in ASME BPVC, Section VIII, Division 3 including the design requirements of Code Cases 2836-3, 2847-1, and 2938-1, and additional design requirements in the User's Design Specification found in SIA report 2100378.401.R0 [9]. Design conditions evaluated herein for the vessel are 500 bar at 93°C (200 ^{0}F). The relief device used for overpressure protection must not allow the pressure in the vessel to exceed the design pressure. The various criteria used to evaluate the design and the outcomes of the various assessments are described briefly and detailed in the Manufacturer's Design Report (MDR) [10].

2.6.1 Global Criteria of Loading

The requirements for protection against global plastic collapse were evaluated for one relevant load combination based on a design pressure of 500 bar at 93°C. This included evaluation of an internal pressure load of 900 bar (1.8 x design pressure) with material properties at 93 ^{0}C. Satisfaction of this criteria was demonstrated through achieving a converged solution in finite element analysis simulating these loading conditions.

2.6.2 Local Criteria of Loading

The criteria for the calculation of local strain damage in ASME VIII-3 was evaluated. ASME VIII-3 requires that local strain damage be evaluated for two loading conditions:

- The single local loading specified in KD-230.4 at 1.28 x design pressure = 640 bar.
- A series of applied loads as described for the evaluation of ratcheting in ASME VIII-3 Article KD234. This series of loads consisted of simulation of the operation of the vessel through three operational cycles as described in Table 2.12.

This calculation methodology was employed at every point in the vessel and the accumulated damage was determined. At no point in the vessel had damage accumulation beyond the accumulation limit of 1.0 established by the ASME VIII-3, thus the design criterion was met.

Table 2.12 – The various loading steps in the analyses of the wrapped cylinder.

Loading Step	Load Descriptions	Applied Pressure, MPa	Temperature ^{0}C
1	Initial Fit Up (reads assembly stresses due to wire pre-tension and autofrettage	0	21.1
2	Hydrostatic Test	66.9	21.1
3	Hydrostatic Pressure Release	0	21.1
4	Operational Pressure/Temperature	50	93.3
5	Pressure Temperature Release	0	21.1
6	Operational Pressure/Temperature	50	93.3
7	Pressure Temperature Release	0	21.1
8	Operational Pressure/Temperature	50	93.3
9	Pressure Temperature Release	0	21.1

2.6.3 Buckling and Service Criteria

There were no design conditions which would result in compressive loading of the vessel that could cause buckling. Therefore, an evaluation for buckling was not required to comply with ASME VIII-3.

2.6.4 Hydrostatic Testing and Ratcheting Criteria

The hydrostatic test criterion was evaluated as part of ratcheting assessment. The first step of the ratcheting assessment is to simulate the hydrostatic test pressure. ASME VIII-3 requires the model to show convergence to comply with the hydrostatic test criteria which was demonstrated.

The remaining series of applied loads described in Table 2.12 were subsequently evaluated for use in evaluation of the ratcheting criteria (ASME VIII-3 Article KD-234). ASME VIII-3 requires that one of the following criteria be met to demonstrate compliance with the ratcheting assessment:

- There is no plastic action (i.e., zero plastic strains incurred)

- There is an elastic core in the primary-load-bearing boundary of the component
- There is not a permanent change in the overall dimensions of the component

The assessment showed that there were no changes in the overall dimensions of the vessel.

2.6.5 Life Assessment of the Vessel

2.6.5.1 Life Assessment of the Wire Wrap

Fatigue analysis of the wire jacket was performed using methods described in KD-9 to ensure that there is no threat of failure of the jacket due to pressure cycling. Further, the stresses in the wire jacket were analyzed at the autofrettage pressure which was about 2.4 x design pressure to demonstrate that the wire jacket remains elastic at the autofrettage pressure. The wire jacket fatigue life was shown to be more than 100,000 cycles.

2.6.5.2 Life Assessment of the Liner

The life assessment of the vessel is in accordance with ASME VIII-3 Article KD-4 of ASME Section VIII Division 3 code. The number of design cycles reported is based on the pressure cycle ranges specified in the User's Design Specification. The number of cycles reported is the total design cycles for the vessel. Actual combinations of cycles and / or pressure should be considered in the equipment operation.

The cyclic life is based on an NDE method that can detect flaws 1 mm deep by 5 mm long at the ID and OD of the vessel, as well as the thread roots. This flaw size was reported as the detection threshold for the NDE methods used in the vessel.

Fracture mechanics-based fatigue crack growth analyses were performed assuming surface flaws that were 1 mm deep and were 5mm long on the surface. The flaw locations for which the analyses were performed are schematically shown in Fig. 2.30 and models used for the flaws at various locations are shown in Table 2.13. As part of this analysis, the cylinder was cycled between the maximum operating pressure (MOP) to 10% of MOP. The MOPs chosen were 450, 410, and 350 bar. Design lives in number of pressurization and depressurization cycles were estimated for the various loading conditions using the method prescribed in ASME Section VIII-Div 3 code article KD-10 that specifically addresses cylinders for hydrogen storage. The results are listed in Table 2.14.

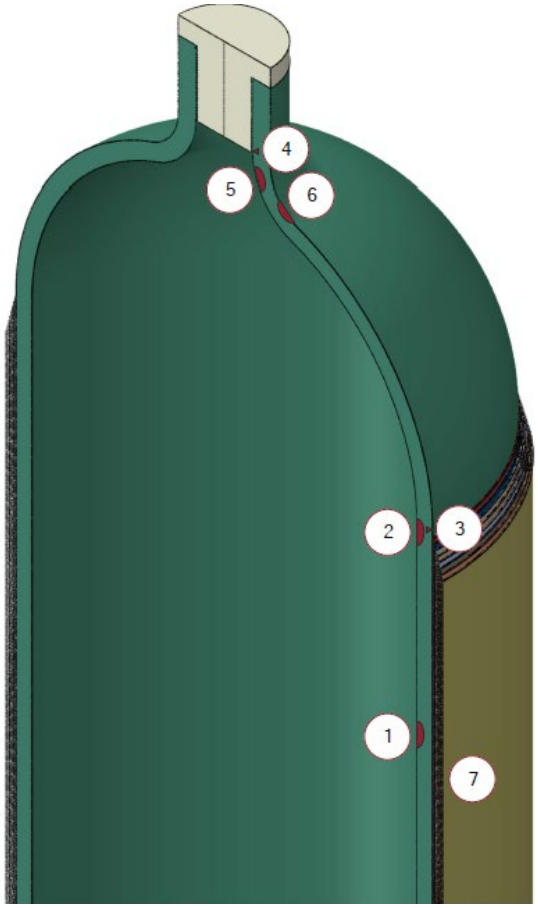


Fig. 2.30 – Locations of assumed flaws in the cylinder for the fatigue crack growth analyses.

Table 2.13 – Crack models used in the fracture mechanics analysis for estimating design life at various points of the cylinder.

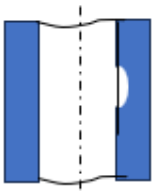
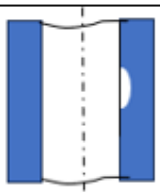
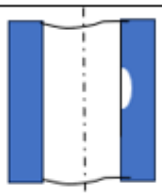
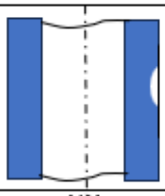
Flaw Evaluated	Crack Model Used			Initial Flaw Size, mm
1	Longitudinal crack at ID of the liner shell		Semi-elliptical longitudinal crack in the inside surface	1mm x 5 mm
2	Longitudinal crack at ID of the liner shell near the end of the wire wrap		Semi-elliptical longitudinal crack in the inside surface	
3	Circumferential crack at OD of the liner shell near the end of the wire wrap		Semi-elliptical circumferential crack on the outside surface	
4	Circumferential crack at the first thread		Semi-elliptical circumferential crack on the inside surface	
5	Longitudinal crack at ID near the top of the vessel (Nozzle Knuckle)		Semi-elliptical longitudinal crack in the inside surface	
6	Longitudinal crack at OD near the top of vessel nozzle knuckle		Semi-elliptical longitudinal crack in the outside surface	
7	Fatigue in wire wrap	N/A	N/A	N/A

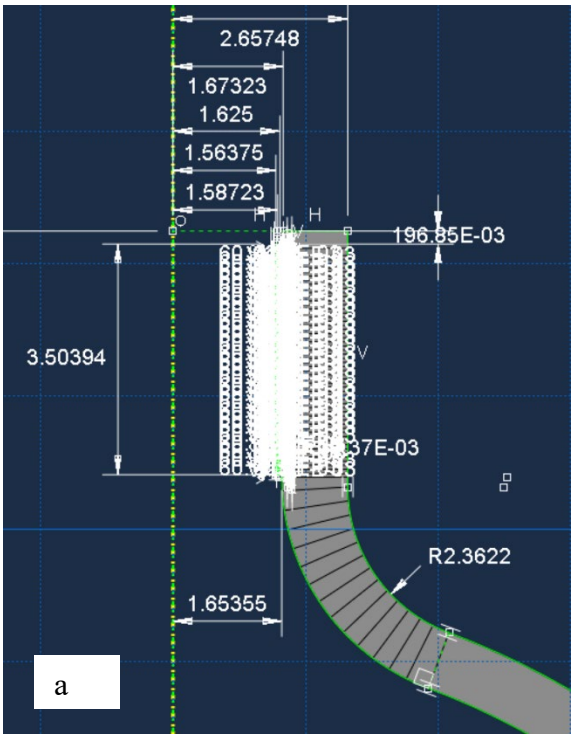
Table 2.14 shows the predicted design life/ inspection intervals that were estimated from these calculations.

Pressure, Bar		Design Life in Cycles, Location					
Maximum	Minimum	1	2	3	4	5	6
350	35	5.58×10^7	13,300	57,000	5,300	3,000	50,000
410	41	122,000	4,450	33,500	1,700	1,600	29,500
450	45	32,000	2,900	26,500	1,100	1,200	23,500

Based on the fatigue lives listed in Table 2.14, the MOPs of 450 bar and 410 bar are ruled out for this design because of insufficient fatigue lives. At 350 bar, the fatigue life exceeds 4000 cycles (200 cycles/year for 20 years) everywhere except for a flaw in the knuckle region where it was calculated as 3,000 cycles or 15 years at 200 cycles/year. This result is worthy of further discussion.

Figure 2.31 shows two finite element analysis models for estimating the stresses in the knuckle region. The model on the left assumes that the wall thickness in the liner equals the minimum specified of 22.9 mm everywhere, while the model on the right assumes that the wall thickness in the dome and the knuckle region is determined by the intersection of extended lines tracing the inner surface of the nozzle with the extension of the curve associated with the inner surface of the dome. This geometry is forced by the forming tools used in forming the domes and are more realistic. The modelled thicknesses are shown in Fig. 2.32b where it is 38.1 mm in the knuckle region. The actual wall thicknesses are shown in Fig. 2.27 to be about 40 mm, or even higher. It thus appears that the model used in Fig. 2.31b is a more realistic but still conservative in representing the liner geometry and dimensions in the knuckle region.

Figure 2.32 shows the stress distributions at location 5 where the cyclic life was 3000 cycles (15 years), Table 2.14, which was the least among all the locations analyzed. The corresponding life for the stress distribution from Model 2 is 6000 cycles, or 30 years. At location 4, the life estimated from Model 2 jumps to more than 15,000 cycles. Thus, location 5 remains the design life controlling location, where the design life is 6,000 cycles. Model 2 has not been incorporated in the MDR but before selling the cylinder, that change can be made. Thus, the design life or inspection interval for the cylinder manufactured is 6,000 pressure cycles (or 30 years @ 200 cycles per year) with amplitudes ranging from an MOP of 350 bar to 35 bar.



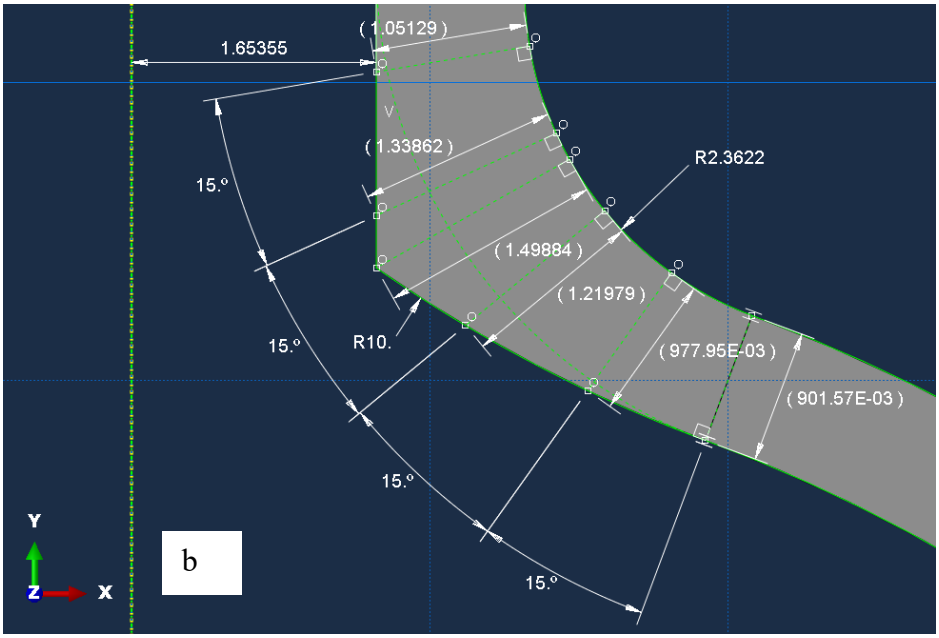


Fig. 2.31 – Two finite element models used to compute stresses in the knuckle region of the liner. In the first model (a) the wall thickness is 22.9 mm throughout the dome and nozzle region and (b) the thickness profile in the dome region as dictated by intersection of the extensions of the inner surface of the liner in the nozzle region and the curvature of the dome at the ID. The wall thicknesses in the various locations are shown in this figure in inches and is as high as 38 mm (1 inch = 25.4 mm).

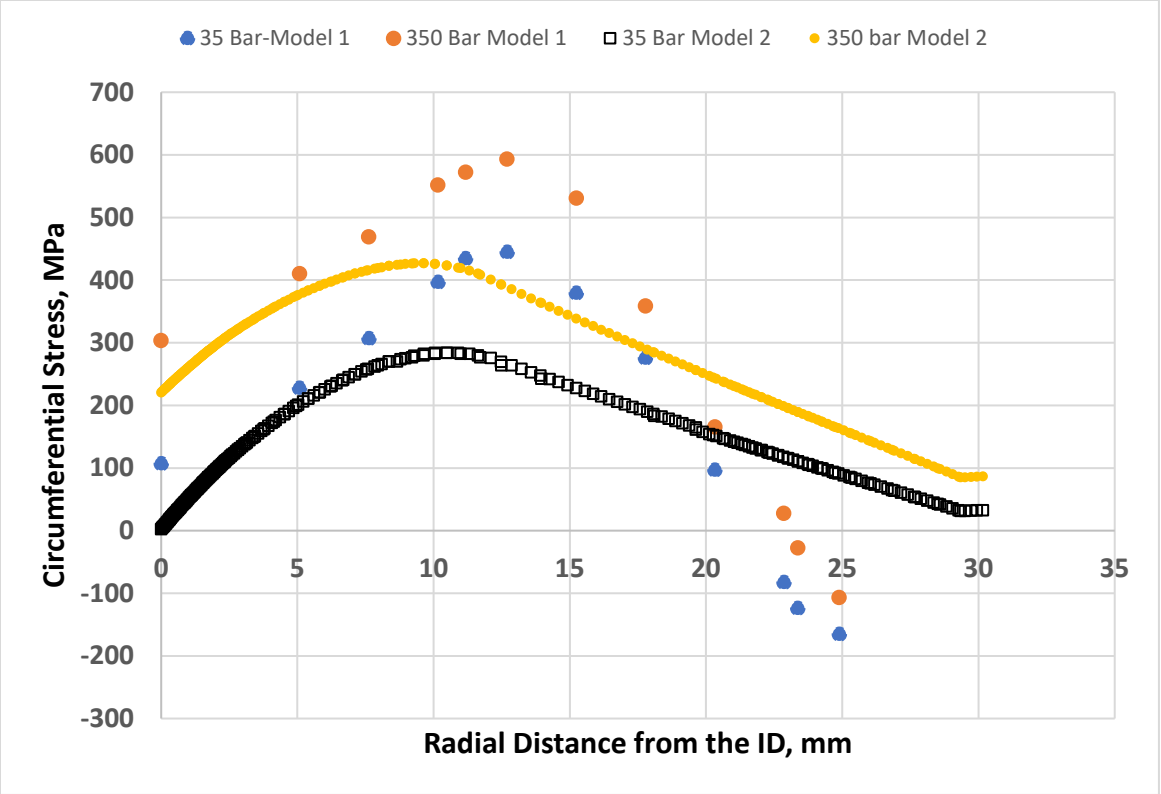


Fig. 2.32 -Estimated stress distributions at location 5 in the knuckle region at maximum and minimum operating pressures using Model 1 that assumes the wall thickness to be the same in the dome and knuckle regions (Fig. 2.31a) and Model 2 using variable wall thickness (Fig. 2.31b) showing significant reduction in stress when Model 2 is used.

2.7 Objective 6: Commercialization and TRL Assessment

WireTough has conceived, designed, produced, and obtained certification for a durable, all-steel Type II-S cylinder constructed using widely available materials for storing bulk hydrogen at high pressures. The cylinder was certified under ASME Section VIII-Division 3 rules for cylinders for ground storage of hydrogen. The cylinder is ready for commercialization in the hydrogen storage market.

The cylinder operates at a maximum operating pressure (MOP) of 350 bar, has a cyclic life of 6000 cycles (30 years at 200 cycles/year) and can store 40.7 KG hydrogen in a single unit (four times that of a Type 1). Further, it is 50% lighter in weight than Type 1 cylinders, has a higher fatigue life and has comparable durability against impact and ballistic loading.

The manufacturing costs of this Type II-S cylinder is estimated at $827/KG of hydrogen stored compared to the price of $1604 per KG of hydrogen stored for Type I cylinders. WireTough has reached the conclusion that the best way to commercialize this product is through partnerships with large scale manufacturers, specially Type 1 cylinder

manufacturer(s), operating under a licensing agreement. This model will allow for the cylinders to be mass produced at multiple locations to meet the high demand. By distributing the licensees across the globe, the logistics costs and time to delivery can also be reduced. A detailed Commercialization Plan is described in Appendix B.

At the beginning of the program, the construction materials for use in Type II-S cylinders to store hydrogen at high pressures were known through research/technology development work performed by Sandia National Laboratory over two decades, and work performed by WireTough as part of a previous DOE sponsored program. WireTough also had previous experience in the design, manufacturing, and commercialization of smaller Type II-S pressure vessels for storing CNG and air using the same construction materials and proprietary manufacturing processes. This justified the TRL-4 level designation at the start of the project. This level was raised to TRL-8 from the work performed as part of this project described in this report in earlier sections. A detailed Technology Maturation Plan is described in Appendix C.

3 Summary and Conclusions

1 WireTough has conceived, designed, and produced a durable, all-steel Type II-S cylinder constructed using widely available materials for storing bulk hydrogen at high pressures. This cylinder was certified under ASME Section VIII-Division 3 rules for cylinders for ground storage of hydrogen. The cylinder is ready for commercialization in the hydrogen storage market.

2 The cylinder operates at a maximum operating pressure (MOP) of 350 bar, has a cyclic life of 6000 cycles (30 years at 200 cycles/year) and can store 40.7 KG hydrogen in a single unit (four times that of a Type 1). Further, it is approximately 50% lighter in weight than Type 1 cylinders, has a higher fatigue life and has comparable durability against impact and ballistic loading.

3 The manufacturing costs of this Type II-S cylinder is estimated at $827/KG of hydrogen stored when produced in batches of 200 compared to the price of $1604 per KG of hydrogen stored for Type I cylinders.

4 WireTough has reached the conclusion that the best way to commercialize this product is through partnerships with large scale manufacturers, specially Type 1 cylinder manufacturer(s), operating under a licensing agreement. This model will allow for the cylinders to be mass produced at multiple locations to meet the high demand. By distributing the licensees across the globe, the logistics costs and time to delivery can also be reduced.

4 References

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Appendix A – User Design Specification (UDS)

December 13, 2022
REPORT NO. 2100378.401
REVISION: 0
PROJECT NO. 2100378.00

Quality Program: ☐ Nuclear ☒ Commercial

Ashok Saxena
WireTough Cylinders, LLC
14570 Industrial Park Road
Bristol, VA 24202

Subject: User's Design Specification

Dear Ashok,

This letter report contains the User's Design Specification (UDS) for the construction of a vessel using WireTough Cylinders wire wound vessel technology.

Please review this thoroughly and let me know if you have any questions or would like to see any changes in the specification.

Very truly yours,



Kannan Subramanian | Associate
Structural Integrity Associates, Inc.®
Powered by Talent and Technology

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Prepared by:		Reviewed by:	
Kannan Subramanian Associate	12/13/2022	Erick R. Ritter Senior Engineer	12/13/2022

User's Design Specification for Hydrogen Gas Storage Cylinder

For

WireTough Cylinders LLC 14570 Industrial Park Road, Suite C Bristol, VA, 24202

According to the rules of the ASME Boiler and Pressure Vessel Code,
Section VIII, Division 3, Alternative Rules for Construction of High-Pressure Vessels,
2021 edition and Code Cases 2836-3, 2847-1, 2938-1

1.0 SCOPE

This specification sets forth design requirements and information as to the intended operating conditions of the pressure vessel being built to this User's Design Specification which WireTough Cylinders is acting as the "designated agent" for the User, as defined in KG-310(a)(3). WireTough Cylinders is acting as an organization that offers pressure vessels for sale or lease for specific service. These requirements and information provide a basis for material selection and design, fabrication, and inspection of the vessel in accordance with the ASME Boiler & Pressure Vessel Code, Section VIII Division 3, 2021 edition and Code Cases 2836-3, 2847-1, and 2938-1. This User's Design Specification covers the requirements pertaining to the definition of equipment and services that shall be considered in assuring Code compliance.

2.0 APPLICABLE CODE

The pressure vessel assembly shall meet all requirements of the rules of the ASME Boiler and Pressure Vessel Code, Section VIII, Division 3, Rules for Construction of Pressure Vessels, Alternative Rules for Construction of High-Pressure Vessels, 2021 edition (VIII-3) and Code Cases 2836-3, 2847-1, and 2938-1. The manufacturer and sub-vendors, if used, shall provide all data using US Customary units of measurement. Dual units are permitted so long as the US Customary units are controlling all stated values, see Code paragraph KG-150.

3.0 VESSEL IDENTIFICATION (KG-311.1)

1. Vessel Number: Each vessel shall be identified by a unique Serial Number by WireTough Cylinders and National Board as each is produced, in accordance with WireTough Cylinders Quality Assurance Manual.
2. Name, Function, and Purpose: The vessels are to be built for stock by WireTough Cylinders, LLC and installed at fixed locations in the 48 continental states of the US for specific service conditions described herein.
3. Service Fluid: The vessel is used for the storage of hydrogen gas.

4.0 VESSEL CONFIGURATION (KG-311.2)

1. Shape: The basic shape of the vessel shall be of a seamless cylindrical shell with swaged heads, wire wrapped on the straight portion of the cylindrical shell, and a threaded connection machined at each end as shown in Figure 1.
2. Orientation: The vessel shall be oriented horizontally in service.
3. Nominal Size or Volume Capacity: The vessels produced will be nominally 1,715L.
4. Support Method and Location: The vessel shall be supported by a supporting mechanism on the OD provided by others. The mechanism will mount the vessel by straps on the OD on the cylindrical portion of the shell over the wire wraps. The weight of the empty vessel is expected to be approximately 9,050 lbs.
5. Functions and Boundaries: The pressure vessel is defined as including the seamless vessel shell and wire wraps including the threaded nozzles on each end.
6. The vessel assembly shall be supplied / shipped with the connection covered or protected by the manufacturer.
7. The end User shall be responsible for installation and certification of the overpressure protection system. The User shall supply the relief device and related components that comprise the relief system. The capacity of the relief device system is not included as part of this User's Design Specification.
8. The end User will ensure that the length of the threads in the pressure relief device are sufficiently long to fully engage all threads in the nozzle of the cylinder. 9. No corrosion or erosion allowance is required.

5.0 CONTROLLING DIMENSIONS (KG-311.3)

WireTough will supply drawings of the vessel. The overall dimensions shall be as follows:

1. Nominal OD = 25.36 inches (644 mm)
2. Nominal Total Length = 299.212 inches (7600 mm)

Vessel pressure connections and openings shall be as follows:

One opening at each end shall be provided, with a 3 ¼ -8UN-2B thread and with the end of the nozzle for the opening machined flat.

6.0 DESIGN CRITERIA (KG-311.4)

The design pressure and design temperature are 7,250 psig and 200 °F. The minimum design metal temperature (MDMT) is -40 °F. No significant thermal gradients are expected across the section while in operation.

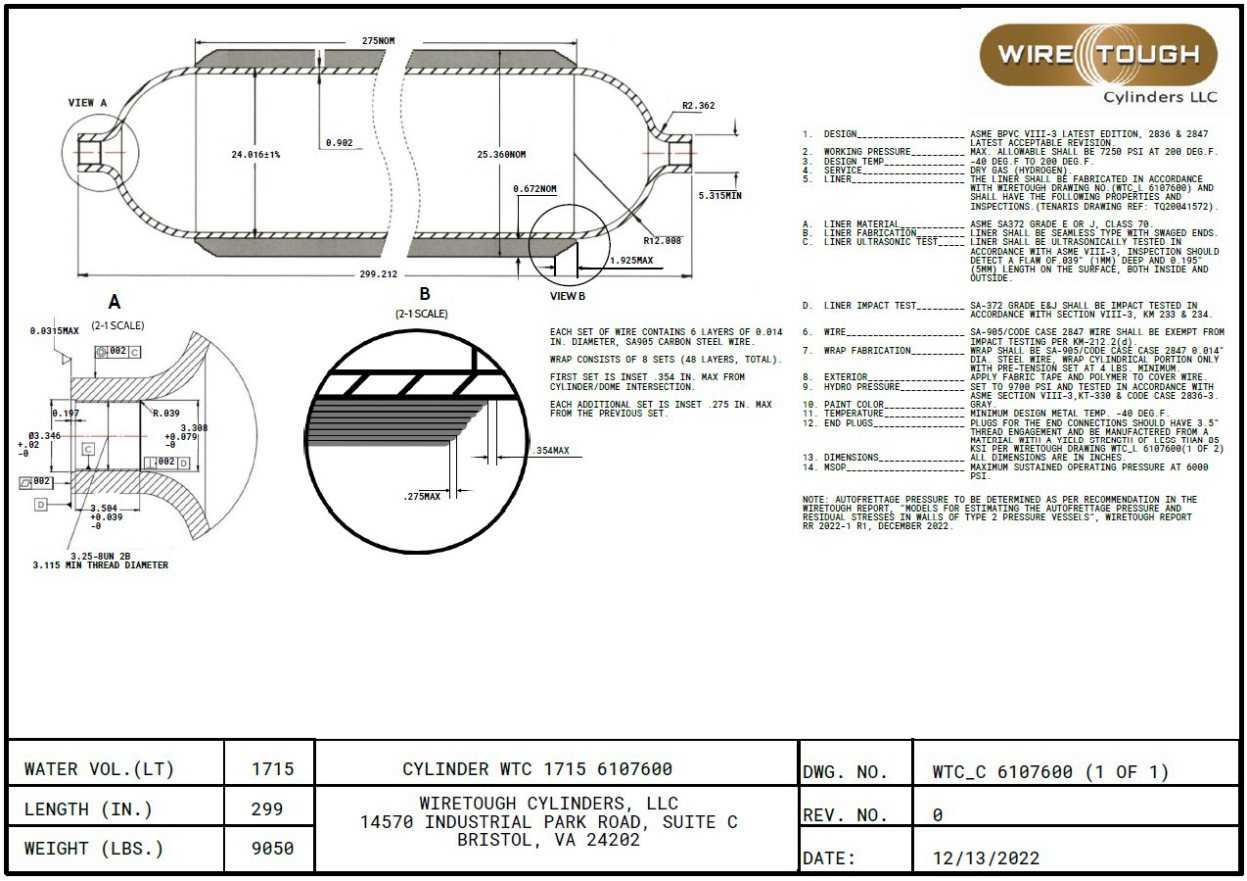


Figure C.1 – Sketch of vessel

7.0 OPERATING CONDITIONS (KG-311.5)

1. Maximum sustained operating pressure (MOP) is 6,000 psig at coincident fluid temperature of up to 200 °F.
2. The pressure vessel is for hydrogen gas storage. No upset conditions are expected.
3. The vessel will not be heated or cooled other than by change in ambient surroundings, or due to filling and venting of products. No transient thermal conditions of significance are expected during operation.
4. The pressure cycles that the vessel will undergo is detailed in Section 11.0.

8.0 CONTAINED FLUID DATA (KG-311.6)

1. The contained fluid phase is gaseous.
2. Density will vary with operating conditions, highest density at highest operating pressure is 1.49 lb/ft^3.
3. The nominal inlet and outlet fluid temperatures range between -40 °F to 200 °F.
4. Jet impingement streams are not applicable.
5. The vessel contains hydrogen which is a flammable, hazardous, and noxious gas.

9.0 MATERIALS SELECTION (KG-311.7)

1. The seamless vessel shell shall be made from SA-372 Grade E or J Class 70 steel. The wire wrap shall be manufactured of a round wire, similar to SA-905, manufactured to meet requirements of Code Case 2847-1.
2. The vessel wire winding is protected from aqueous contact from the ambient environment by means of a protective outer covering of fiberglass and epoxy. This coating is not part of the structural support for the shell.

10.0 LOADINGS (KG-311.8)

1. No external loadings are to be considered other than pressure reactionary forces.
2. There will be no loading due to jetting or impingement.
3. The vessel will not carry the weight of any external components.
4. See Section 7.0 for operating conditions.
5. Evaluation of snow loadings are not required.
6. All applicable loadings that are listed in paragraph KD-110(i) shall be considered.

   Note: KD-110(i) states “residual stresses, introduced at fabrication, e.g., by autofrettage, hydrostatic test, shrink fit, prestressed wire or strip winding, rolling, forming, welding, thermal treatments, and surface treatment such as shot peening”
7. The vessel contains a gas, and the weight should be considered as negligible.
8. The weight of the vessel itself should be considered in the design. The vessel is to be supported by two external clamps.
9. The worst-case wind and earthquake loading for the continental United States should be considered.

11.0 OPERATING LIFE (KG-311.9) AND FATIGUE ANALYSIS (KG-311.10)

1. The cyclic life of the vessel is to be determined analytically during the design process.
2. This calculated cyclic life shall be used by the User as the basis for determining recommended vessel inspection frequency and recommended inspection documentation.
3. The vessel cyclic life is to be based on 20 years of service with number pressure cycles permitted for the following operating pressures:
    a. 6,500 psi to 650 psi
    b. 6,000 psi to 600 psi
    c. 5,500 psi to 550 psi
    d. 5,000 psi to 500 psi

12.0 OVERPRESSURE PROTECTION (KG-311.11)

The end User shall be responsible for installation and certification of the overpressure protection system. The User shall supply the relief device and related components that comprise the relief vent system. The calculations, test reports, and other information used to justify the size, location, connection details, and flow capacity of the relief system is not included as part of this User's Design Specification.

13.0 ADDITIONAL REQUIREMENTS (KG-311.12)

1. U.S. Customary units shall be used by the manufacturer and sub-vendors, if used, in all certified documents, and all markings, and stamping required by the code. Dual units are permitted so long as the US Customary units are controlling all stated values, see Code paragraph KG-150.
2. The completed vessel shall be hydrostatically tested at room temperature in the horizontal position at the Manufacturer's facility.
3. The vessel shall be tested with water. The water temperature shall be between 55°F and 80°F. Test pressure shall not be applied until the vessel and the fluid are within 10°F of each other.
4. The vessel interior shall be cleaned and dried as soon as practical after the hydrostatic test and purged with inert gas such as nitrogen to prevent corrosion during shipment and storage. The exterior of the vessel, the supporting mechanism, and the associated components shall be protected by a suitable coating to prevent rust during shipment, installation, and operation of the vessel.

5. The Manufacturer shall furnish to the User the records listed below as a minimum:
   a. User's Design Specification complete with all reference documents (KG-311)
   b. Manufacturer's Data Report (KS-300)
   c. Manufacturer's Design Report (KG-323)
   d. Manufacturer's Construction Records:
      i. As-built drawings
      ii. Materials certifications
      iii. Material Test Reports iv. Record of repairs if repairs were done
   e. NDE Examination Procedures, Records of Procedure Qualification, and Records of Personnel Qualifications (KE-105)
   f. NDE reports and as-built sketches showing all relevant NDE indications
   g. Record of non-conformances and disposition. In addition, the Manufacturer shall maintain the required records and a copy of all the above data for a period of not less than 5 years. These records shall be made available upon request to AI Supervisors or review teams designated by the ASME.
   h. Record of all heat treatments (these records may be either the actual heat treatment charts or a certified summary description of heat treatment time and temperature)
   i. Vessel Production Test Reports including Hydrostatic Testing and Volumetric Expansion Testing in accordance with ASME VIII-3 and Code Case 2836-3.

6. The requirements of KD-10 shall be met including the requirements of the proposed Code Case 2938-1.

14.0 INSTALLATION SITE (KG-311.13)

The pressure vessel system is to be built for stock and installed after purchase by the final User at a fixed location in the continental US.

Appendix A Certification of the User's Design Specification

CERTIFICATION OF COMPLIANCE OF THE USER'S DESIGN SPECIFICATION

Signature Indicating Concurrence of WireTough LLC Representative

I, the undersigned, being experienced and competent in the field of design related to high pressure vessels relative to this User's Design Specification, hereby certify that to the best of my knowledge and belief it is correct and complete with respect to the Design and Service Conditions given and provides a complete basis for construction in accordance with Article KG-3 and other applicable requirements of the ASME Boiler and Pressure Code, Section VIII, Division 3, Alternative Rules for Construction of High Pressure Vessels, 2021 Edition and Code Cases 2836-3, 2847-1, and 2938-1 and is complete except for the inclusion of the relief device system to KG-311.11. This certification is made on behalf of WireTough Cylinders acting as the designated agent in accordance with KG-310(a)(3):

Certified by:

Kannan Subramanian, PhD, PE
North Carolina Professional Engineer License Number 036898

Contact Data:
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Areas of responsibility:
All except pressure relief device system

Professional Engineer Seal:



Date: 12/13/2022

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Appendix B – Commercialization Plan

Submitted To: U.S. DOE/NETL - National Energy Tech Lab
DOE Award Number: DE-FE0032033
Project Title: Durable Low-Cost Pressure Vessels for Bulk Hydrogen Storage
DOE Project Manager: Vito N. Cedro, Project Manager
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Date of Report: August 28, 2023
UEI Number: JDRLX4PK3BUS
Recipient: WireTough Cylinders LLC, Bristol, VA 24202
Project Period: 3/1/2021 – 8/31/2023
Reporting Period Date: 3/1/2021 – 8/31/2023

Report Term or Frequency: One Time deliverable

1. Summary

WireTough Cylinders has successfully designed, produced, and has successfully tested and obtained ASME certification for its fatigue resistant and durable Type II-S pressure vessel that can economically and safely store 40.7 KG of hydrogen at a maximum operating pressure of 350 bar in a single cylinder. This vessel is now ready for production and commercial deployment. Preliminary designs have also been completed for vessels that are rated to operate at pressures of 500 and 700 bar that can store 42 and 29.5 KG hydrogen, respectively. Compared to Type 1 cylinders, WireTough's solution offers a 55 to 62.5% weight savings for every KG of H_2 stored and can store four times the hydrogen in a single vessel compared to a comparably sized Type 1 vessel. The manufacturing cost of WireTough's Type II-S cylinder is estimated to be $827 per KG of hydrogen stored, when mass produced in quantities of 200/year or more. For comparison, the price of a Type 1 cylinder is estimated to be $1604 per KG of hydrogen stored.

As part of its commercialization plan, WireTough is seeking partners among large manufacturers of pressure vessels to commercially offer these vessels through licensing agreements. WireTough is also preparing an application for the DOE's MAKEIT program to build its own facilities for manufacturing up to 200 vessels per year and offer them commercially. In the past two years, WireTough has received 71 inquiries for high pressure hydrogen cylinders that add up to potential sales revenues of $500 M, indicating a strong demand for its product.

2. Overview of Hydrogen Demand and Demand for Storage Solutions

As of April 2022, 131 countries around the globe had announced net zero emissions target to be met by the year 2050.[1] The need for clean sustainable energy with long-term availability and cost effectiveness is a clear global priority. United States has made a massive investment of $8 billion through the 2022 Infrastructure Bill. This "Earth Shot" will focus on accelerating the required hydrogen infrastructure through four or more regional hubs. These regional hubs are designed to take full advantage of the utilitarian nature of hydrogen. The success of this initiative is strongly dependent on bulk hydrogen storage capability.

Green hydrogen provides a solution to many of today's clean energy needs as well as a bridge to new technologies of the future. As of 2021, approximately 75 megatons of pure hydrogen and an additional 45 megatons of hydrogen in the form of mixed gases are being produced annually.[2] It is expected that by 2050, the demand for hydrogen will grow to seven times over its current demand, reaching upward of 530 megatons[2]. The overall hydrogen economy is expected to have annual revenues of $2.5 trillion by 2050 while continuing to be in a strong growth phase.[3]

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[1]https://www.irena.org/Energy-Transition/Technology/Hydrogen#:~:text=Hydrogen%20is%20produced%20on%20a,of%20a%20mix%20of%20gases.

[2] IEA (2022), *Global Hydrogen Review 2022,* IEA, Paris; https://www.iea.org/reports/global-hydrogen-review-2022, License: CC BY 4.0

[3]"Hydrogen Scaling up: A Sustainable Pathway for the Global Energy Transition." Hydrogen Council, hydrogencouncil.com/en/study-hydrogen-scaling-up/

Hydrogen has several uses and is often referred to as the Swiss Army Knife of energy. These applications include,

- Electricity generation by acting as a heat source,
- As an energy storage medium with a long shelf-life for storing excess power generated in fossil-fuel, nuclear, and renewable power generating stations,
- Fuel for back-up power for large buildings and businesses or in areas affected by natural disasters,
- Fuel for all types of vehicles used in ground, air, and marine transportation,
- In production of chemicals and raw materials.

Even though hydrogen is the most abundantly available element on earth in the form of water, it has been difficult and expensive in the past to harness it as an energy source because of its low volumetric energy density under ambient conditions (measured in Joules/m^3). Thus, a high level of compression is necessary to compete favorably with liquid fossil fuels such as gasoline. Also, hydrogen causes embrittlement in steels. making storage at high pressures in gaseous form challenging. Recent improvements in pressure vessel technology have addressed these challenges directly and WireTough's technology is an example of such advances[4].

3. Demand for Hydrogen Storage Cylinders

With predicted rapid growth in the demand for hydrogen, storage tanks (or cylinders) will also be in high demand. Having storage vessels with various pressure ratings that are durable against fatigue loading, safe, and cost-effective are essential to meeting these needs. The need for hydrogen storage is not limited to only large storage depots; it also extends to storage in refueling stations, fossil, nuclear, and renewable energy generating plants and many others. WireTough's platform technology embodied in the Type II-S pressure vessels consisting of one-piece steel liners that are wrapped with several layers of a fine, ultra- high strength steel wire provides significant cost, weight, and space advantages over Type 1 pressure vessels.

By 2030, it is expected that more than 15,000 new hydrogen refueling stations will be built worldwide; this will generate a need for an estimated 150,000 tanks with each capable of storing between 30 to 50 KG of hydrogen. Further, with 250 to 300 TWH (Terra-watt-hour) of surplus renewable electricity available for storage, there will be a need for hundreds of thousands of tanks, even if only 10% of this surplus energy is stored in the form of gaseous hydrogen. Hydrogen offers advantages over batteries in energy storage. For example, energy storage in rechargeable batteries has a shelf-life of only a few days, whereas energy stored as hydrogen can be accessed over months making it a viable medium for long duration energy storage (LDES). Batteries have limited lives while hydrogen storage systems can be designed to last 30 years, and more.

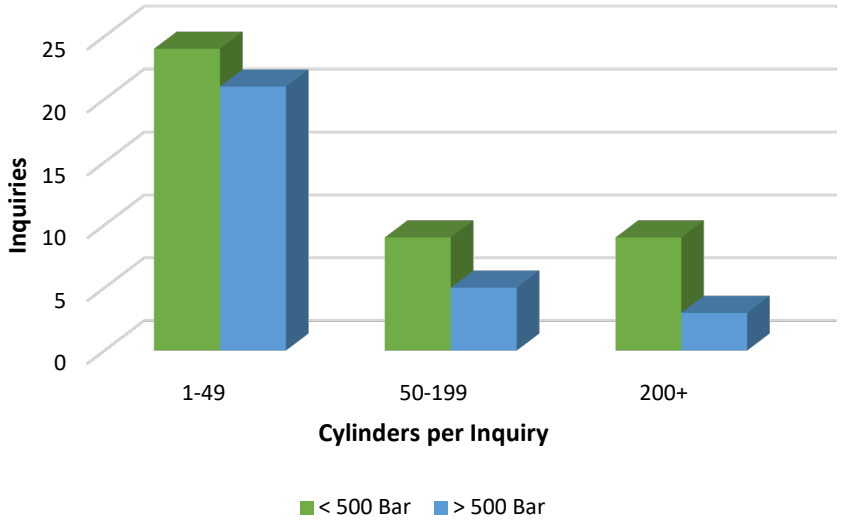
The projected annual market for ground storage tanks is expected to be $100 B during the next 10 years. In addition, an estimated 750,000 fuel tanks will be needed for onboard vehicle storage in off-road vehicles, construction equipment, and mid-size trucks, where weight is not as important a consideration as in passenger vehicles.

[4] A. Saxena, et al. "Durable Low-cost Pressure Vessels for Bulk Hydrogen Storage" Final Contract Report, DE-FE-0032033, September 2023, WireTough LLC

In the past two years, WireTough has received **71 inquiries for supplying hydrogen storage tanks for ground storage facilities** that range in capacity from 50 KG to several hundred thousand KGs of hydrogen. The specified hydrogen pressure ranged from 350 to 700 bar. A total of 41 of those inquiries were for hydrogen storage between 350 to 500 bar and the remainder for between greater than 500 to 700 bar. The upper limit of 700 bar is dictated by limitations of compressor technologies and energy spent in compression. A categorization of the various inquiries is shown in Fig.1 and is described below:

- 45 of the 71 inquiries were for up to 49 cylinders to store hydrogen with each cylinder storing 40 KG of hydrogen.
- Nine of 71 inquiries required 50 to 199 cylinders. These inquiries were for future hydrogen refueling stations and for large, bulk hydrogen storage depots. Typically, hydrogen refueling stations were looking for five to eight units per station, each capable of storing 40 to 50 KG of hydrogen per cylinder.
- Nine inquiries received were seeking over 200 cylinders each. These were for large projects requiring massive hydrogen storage for use in both energy storage/generation and for creation of hydrogen-based alternative fuels, such as liquid ammonia. Four of these projects were seeking more than 500 cylinders to be produced and delivered within a two-year period.

If the 71 inquiries turned into actual sales, it is estimated to yield $500M in revenues. It is also worth noting that these inquiries have been generated purely based on the descriptions provided on WireTough's website and no active marketing efforts. Thus, these market size estimates are most likely conservative.



Fig, 1 – Commercial inquiries for hydrogen storage cylinders received by WireTough Cylinders during the period of 03/2020 to 07/2023.

4. WireTough's High Pressure Gaseous Hydrogen Storage Solution

WireTough has engaged in federal, and state (Virginia) sponsored research and has developed the technology for three complementing models of cylinders suitable for storing hydrogen at high

pressures. The status of the three cylinders is summarized in Table 1. The technical specifications prepared during design of these cylinders were guided by the inquiries that WireTough has received. These cylinders can collectively meet the technical requirements of the various inquiries described in the previous section. The current TRL levels of the various cylinders are also noted in Table 1. Figure 2 shows (a) pictures of liners used for building the 350 bar, 1715 Liter cylinders, (b) the cylinder being wire-wound and (c) picture of an as wound cylinder ready for autofrettage and hydrotesting. Figure 3 shows a built 700 bar, 765 L vessel capable of storing 29.5 KG of hydrogen. The latter was built under a prior DOE contract.

Table 1 – Status of the proposed three models of cylinders for storing hydrogen.

Cylinder Description			TRL Level	Status	Steps remaining for commercialization
Vol, L	MOP* bar	MAWP* bar			
1715	350	50	8**	• Prototype built, • Autofrettaged and pressure tested, • A valid MDR prepared, • ASME certification being currently pursued and expected by 9/30/2023.	• Ready for commercialization after certification
765	700	1000	5***	• Prototype built on a previous contract to demonstrate design and manufacturability but not yet autofrettaged and hydrotested.	• MDR needs updating, • Build a new prototype, • Obtain ASME certification
1400	500	650	5***	• Detailed design and analysis completed	• MDR needs preparation, • Needs a prototype, • ASME certification needed.

*MOP = Maximum operating pressure; MWAP = Maximum allowable or design pressure.

** This TRL level is contingent upon successfully completing the autofrettage and hydrotest scheduled on the cylinders in September of 2023.

***These TRL levels can very rapidly be moved to level 8 because of the design and manufacturing procedures developed as part of this program that led to the 350 MOP cylinder.

The materials of construction for these cylinders are all commercially available and in most cases are off-the-shelf items, listed in Table 2. The only special purpose item is the liner, but even that is built from a steel pipe of commercially available, standard size using a steel that has been used in pressure vessels for several decades. There is sufficient commonality between the manufacturing processes for cylinders of various sizes and pressure ratings so success in building any one cylinder provides confidence that the others can also be built. For example, the success in building a prototype of the 1715 L - 350 bar cylinder and 765 L, 700 bar cylinders gives confidence that the other models can also be built. The weight of the liners and the number of wire layers, as

well as the wrapping process and equipment are similar for cylinders of various sizes in Table 1. Figure 4 shows the costs involved in building these vessels for an annual production volume (APV) of 200. Cost of materials and in particular, the cost of the liner, stands out. Therefore, a stable supply of liners is crucial to producing Type II-S cylinders.

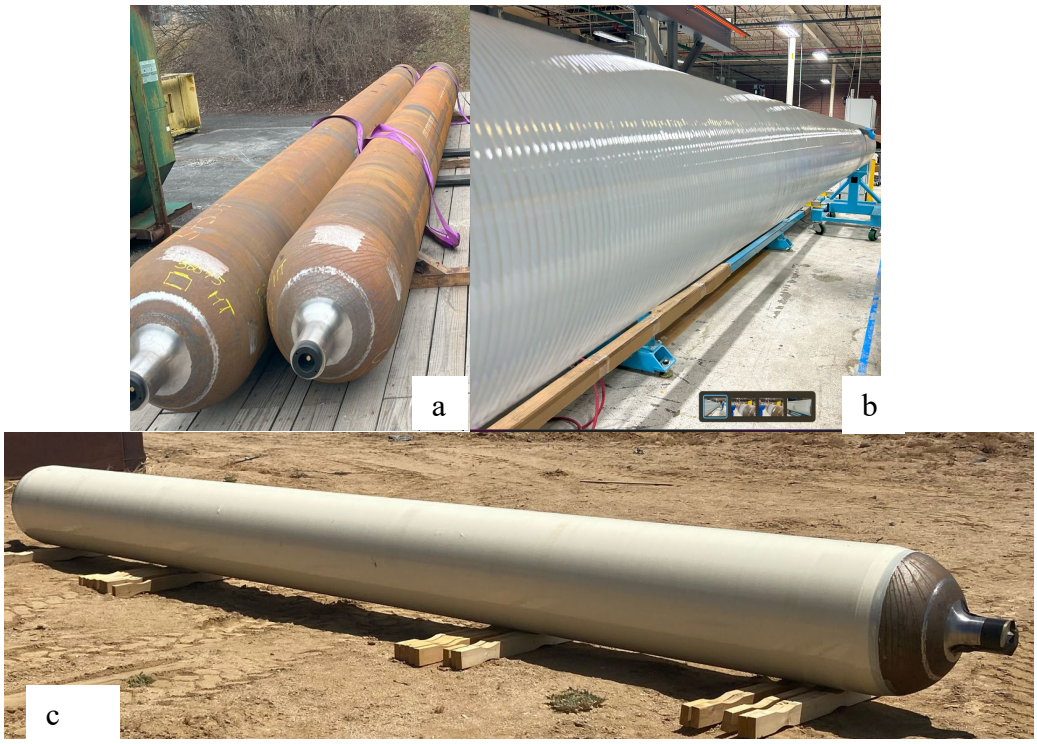


Fig. 2 – (a) Picture of the two seamless liners being delivered to WireTough's facility (b) picture of the cylinder during wire winding (c) a 350 bar MOP, 650 mm OD, 1715 L capacity cylinder capable of storing 40.7 KG of hydrogen after wire-winding and UV coating, ready for autofrettage and hydro-testing.

Table 2– List of materials of construction for manufacturing Type II-S wire wrapped vessel.

Item	Commercial Source
Liner (SA272 Grade E/J – Class 70 -Pressure Vessel Steel	Tenaris Dalmine SPA, TX CPI McKeesport, PA
SA 905 0.35 mm diameter round Wire	Kiswire, Japan with office in Pine Bluff, Arkansas
Epoxy resin from 3M	3 M, Minneapolis, MN
UV Protective skin	Pittsburgh Paint and Glass, Pittsburgh, PA



Fig. 3 – Picture of a 700 bar MOP, 440 mm OD cylinder after wire wrapping.

5. Commercialization Strategy

Full-scale prototypes and fully operational 350 bar, 1715 L cylinders were successfully designed, produced, tested, and are in the process of certification as part of this project[5]. This cylinder will be ready to be offered commercially after certification. Figure 4 shows how the cost of hydrogen storage as represented by $/per KG of hydrogen stored varies with annual production volume (APV) from 20 cylinders per year to 200 cylinders per year. The cost begins to level off at approximately 150 cylinders per year. A single manufacturing line working in two-shifts can produce up to 200 cylinders per year. At an APV of 200, the storage cost reduces to $827 per KG of H_2 stored. For comparison, the price of Type 1 cylinders is $1604 per KG of H_2 stored. Similar calculations were performed for cylinders with MOP of 500 and 700 bar and are shown in Fig. 5. The weight of the Type II-S cylinder is 90.3 KG per KG of H_2 stored compared to the weight of Type 1 which is 200.6 KG per KG of H_2 stored, a 55% decrease in weight.

Figure 6 shows the various elements of manufacturing costs of a Type II-S cylinders. With the cost of steel (liner and the wire) being the most important component of the cylinder cost, the cost-differential between Type 1 and Type II-S is understandable. Further, it will take 4 Type 1 cylinders to store the amount of H_2 stored in a single Type II-S cylinder. Thus, the footprint of a storage facility utilizing Type II-S cylinders will be approximately four times smaller than the one using Type 1 cylinders for the same quantity of stored H_2 (by weight). Table 3 shows the differences in cylinder weight per KG of H_2 stored between Type I and Type II-S at various MOPs.

[5] This assumes that the autofrettage and hydrotest scheduled in September of 2023 will be successful.

The advantages of switching from Type 1 to Type II-S cylinders are many and make a compelling case for their use over Type 1 cylinders.

Strategies for creating a manufacturing line for producing Type II-S cylinders in quantities that make them economical are discussed next. A single production line capable of producing up to 200 cylinders per year will require the following capital equipment at an estimated cost of $3M for procurement of the equipment and installation.

- A 2-axis winder capable of wire winding cylinders of up to 750 mm in OD, 10 m long and weigh up to 5500 KG,
- Overhead cranes for handling the cylinders and creels for feeding the wire during manufacturing,
- An onsite hydrotest facility for autofrettage and hydrotesting cylinders with pressure ratings of up to 300 MPa (40,000 psi), and associated handling equipment for cylinders that are 750 mm in outside diameter and 10 m long.
- The estimated space needed for a single line of manufacturing and storage will be approximately 1400 square meters or 15,000 square ft.

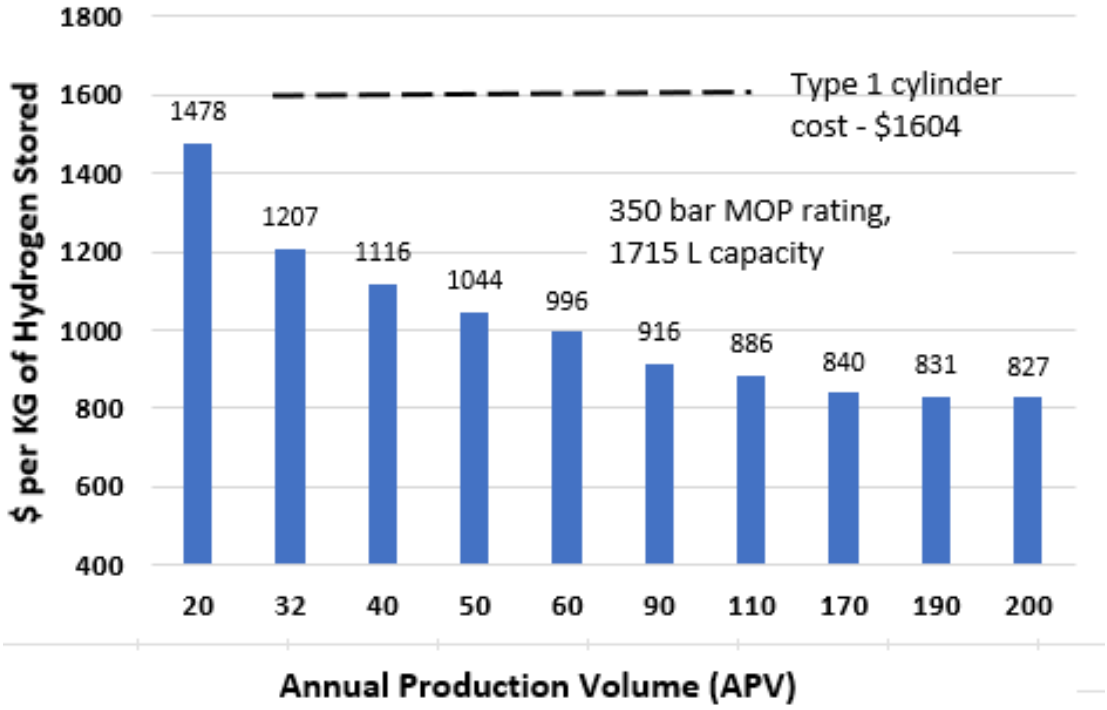
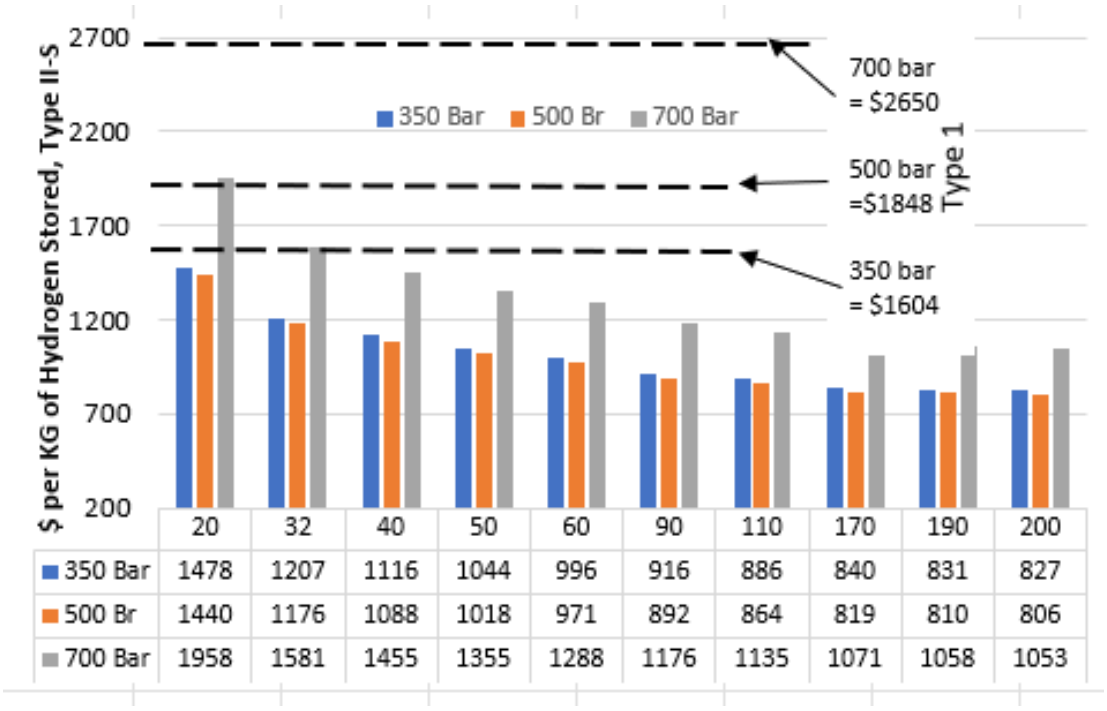


Fig. 4 - Cost of 350 bar MOP. 1715 L liner as a function of annual production volume (APV) in a single production line.



	20	32	40	50	60	90	110	170	190	200
350 Bar	1478	1207	1116	1044	996	916	886	840	831	827
500 Br	1440	1176	1088	1018	971	892	864	819	810	806
700 Bar	1958	1581	1455	1355	1288	1176	1135	1071	1058	1053

Fig. 5 – Comparison of per KG H_2 storage costs as a function of APV for Type II-S and Type 1 cylinders for storage pressures of 350, 500, and 700 bar.

Table 3 – Comparison of cylinder weights per KG of H_2 stored between Type II-S and Type 1 cylinders as a function of MOP.

MOP, bar	Max OD, mm		Capacity, L		KG of H_2 stored in one cylinder		Cylinder weight, KG per KG of H_2 Stored	
	Type II - S	Type 1	Type II-S	Type1	Type II-S	Type 1	Type II-S	Type 1
350	650	356	1760	444	40.7	10.3	90.3	200.6
500	546	280	1400	276	42.0	8.37	93.5	231.0
700	445	203	785	108	29.5	4.08	122.1	325.4

WireTough is primarily an R&D company with facilities for prototyping and small-scale manufacturing and is not equipped to meet the high demand for cylinders; it is thus best to make the technology available for licensing to large-scale manufacturers. Therefore, the proposed commercialization strategy is to find licensing partners who can produce the cylinders on a large-scale. In addition, WireTough will compete in DOE's MAKEIT Prize competition with a proposal to build a facility at its current location that can produce 200 cylinders per year, in addition to pursuing licensing opportunities. The ideal licensee candidates are manufacturers of large liners and Type 1-cylinder manufacturers used in the construction of these cylinders. Figure 6 shows the various elements of the manufacturing costs that are dominated by materials costs. The costs are shown for cylinders with pressure ratings of 350, 500, and 700 bar. The overall costs expressed as \$/KG of H_2 stored decrease only marginally with increasing MOP values. Thus, the choice of MOP can be based on other factors such as compression capability and the associated costs, available floor space, and the end uses of the hydrogen being stored. It is also important that

manufacturing facilities to be globally distributed to serve the needs of their respective regions and thereby minimize shipping costs and delivery times. WireTough can then focus on:

- Identifying partners and signing agreements for transfer of technology,
- training personnel as part of technology transfer,
- Assist in complying with ASME code requirements,
- Continue working on product improvements, new product development, and keeping certifications current.

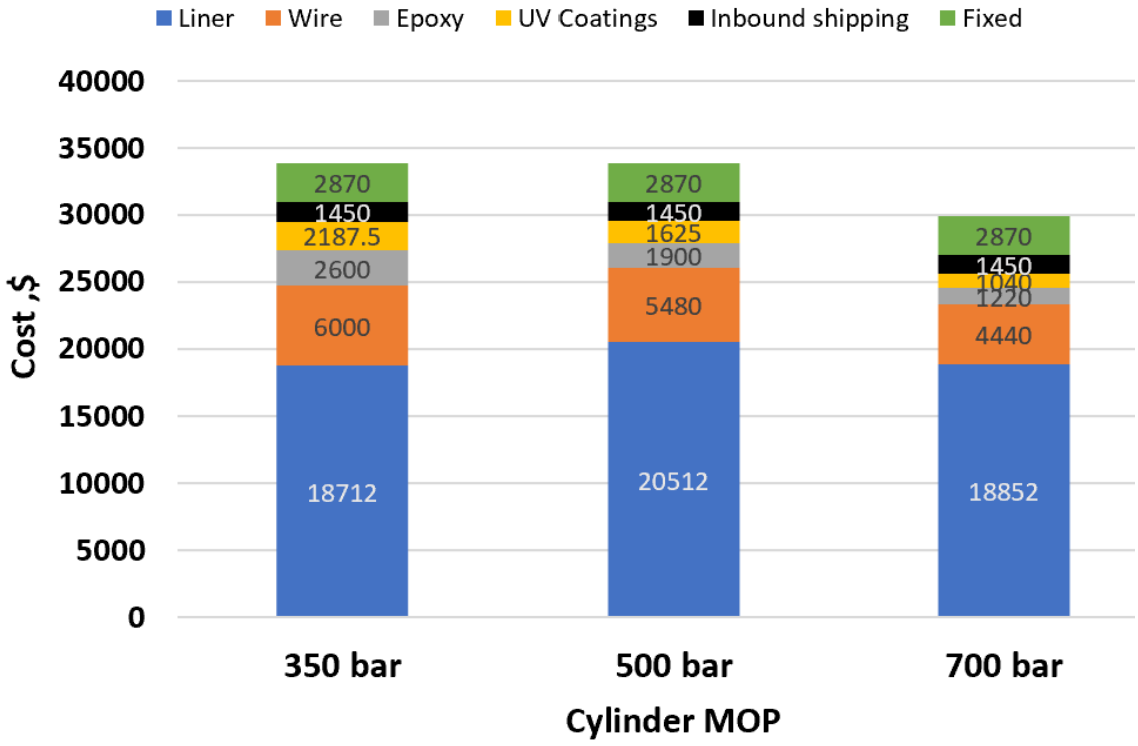


Fig. 6 – Manufacturing cost elements of Type II-S cylinders for ground storage of H_2 at high pressures, assuming an annual production volume of 200 cylinders.

6. Summary and Conclusions

- WireTough has conceived, designed, and produced a durable, all-steel Type II-S cylinder constructed using widely available materials for storing bulk hydrogen at high pressures. This cylinder has been fully certified under ASME Section VIII-Division 3 rules for cylinders for ground storage of hydrogen.
- The cylinder operates at a maximum operating pressure (MOP) of 350 bar, has a cyclic life of 6000 cycles (30 years at 200 cycles/year) and can store 40.7 KG hydrogen in a single unit (four times that of a Type 1). Further, it is 50% lighter in weight than Type 1 cylinders, has a higher fatigue life and has comparable durability against impact and ballistic loading.
- The manufacturing costs of this Type II-S cylinder is estimated at $827/KG of hydrogen stored compared to the price of $1604 per KG of hydrogen stored for Type I cylinders.

- WireTough has reached the conclusion that the best way to commercialize this product is through partnerships with large scale manufacturers, specially Type 1 cylinder manufacturer(s), operating under a licensing agreement. This model will allow for the cylinders to be mass produced at multiple locations to meet the high demand. By distributing the licensees across the globe, the logistics costs and time to delivery can also be reduced. I addition, WireTough intends to compete in DOE's MAKEIT prize program. If successful. WireTough will build a facility to produce 200 cylinders per year in addition to licensing technology to other large-scale manufacturers.

Appendix C – Technology Maturation Plan

A. TECHNOLOGY READINESS LEVEL

The objective of the project was to design and produce a full-scale prototype of a pressure vessel for storing 40 to 50 kilograms of gaseous hydrogen at an operating pressure of 350 to 500 bar (6,525 to 7,250 psi). The goal was to design the pressure vessel as per applicable ASME design codes, produce a full-scale prototype of the vessel to demonstrate the manufacturing processes, and test the vessel to demonstrate full compliance with ASME Section VIII- Division 3 rules for certification. WireTough's assessment of the technology readiness level at the start of the program was TRL-4 as explained below.

At the beginning of the program, the construction materials for use in Type II-S cylinders to store hydrogen at high pressures were known through research/technology development work performed by Sandia National Laboratory over two decades that are incorporated in ASME Design Codes, and work performed by WireTough as part of a previous DOE sponsored program[6]. WireTough also had previous experience in the design, manufacturing, and commercialization of smaller Type II-S pressure vessels for storing CNG and air using the same construction materials and proprietary manufacturing processes. This justified the TRL-4 level designation at the start of the project. This level was raised to TRL-8 with the work performed as part of this project.

B. WORK COMPLETED DURING THE PROJECT

WireTough's approach for designing pressure vessels for hydrogen storage at high pressures consists of using conventional steel liners and wrapping them with commercially available ultra-high strength, 0.35mm (0.014") diameter steel wires using its proprietary wrapping processes. The steel liner selected for this project was 610 mm (24") OD and 7600 mm (25.5') long, with a water capacity of 1715 L. The minimum wall thickness of the liner was 22.9 mm, and it weighs approximately 2700 KG. During this project, the following work was completed to justify the new TRL rating:

Liners made from SA 372 Grade E Class 70 steel approved for use by ASME design codes in high pressure hydrogen storage applications were procured for the project as per specifications prepared by WireTough. The liners were made from seamless tubes with swaged ends and contained no welds, an ASME code requirement. See Fig. 1 for a picture of the liners procured.

WireTough used its prior experience in designing and prototyping pressure vessels to develop a preliminary design for the cylinder as part of this project. The design was evaluated using state-of-the-art finite element analyses, advanced design optimization software tools, and fracture mechanics analyses by the Siemens Corporate Technology Center, Charlotte, NC.

[6] A. Saxena et al., "Low-Cost Hydrogen Storage at 875 Bar Using Steel Liner and Steel Wire Wrap", Final Contract Report, DE-EE-0006668, December 2018.



Fig. 1 – Picture of the liners used for manufacturing the wire-wound cylinders.

The stresses generated on the liner during wire winding and pressurization of the cylinder during autofrettage were estimated by finite element analyses. The autofrettage pressure chosen was sufficient to deform the liner plastically while the wire jacket remained elastic. Upon releasing the autofrettage pressure, the wire jacket exerted high residual compressive hoop stresses on the liner walls, Fig. 2. Subsequently, the cylinder was pressurized to the maximum operating pressure (MOP) to estimate the cyclic stresses during service. These cyclic stresses were used to estimate the design life of the vessel using fracture mechanics. The design was tested against all possible failure criteria required under ASME Section VIII- Division 3 design codes. From these evaluations, it was shown that cylinder can hold 40.7 KG of H2 at a maximum operating pressure of 350 bar and can be safely cycled for 6000 cycles between MOP and the 10% of MOP prior to re-inspection. At a rate of 200 cycles per year, this will take 30 years. The critical region of the cylinder that limits its cyclic life is the exposed area of the liner in the dome region and is accessible for ultrasonic inspection for extending its service life beyond 6000 cycles.

A user design specification (UDS) and a manufacturer's design report (MDR) were prepared by Structural Integrity Associates (SIA) who independently verified the integrity of the design and certified it using design criteria and rules established under ASME Section VIII-Division 3 design code. This cleared the way for manufacturing two full-scale prototype vessels.

The liners were wrapped with several layers of steel wire using WireTough's proprietary processes following procedures documented in WireTough's Quality Assurance Manual (proprietary) and the MDR (proprietary) prepared by SIA. A full-scale experimental model of the cylinder was produced in WireTough's Bristol facility, see Fig. 3 for a picture of wrapped cylinder.

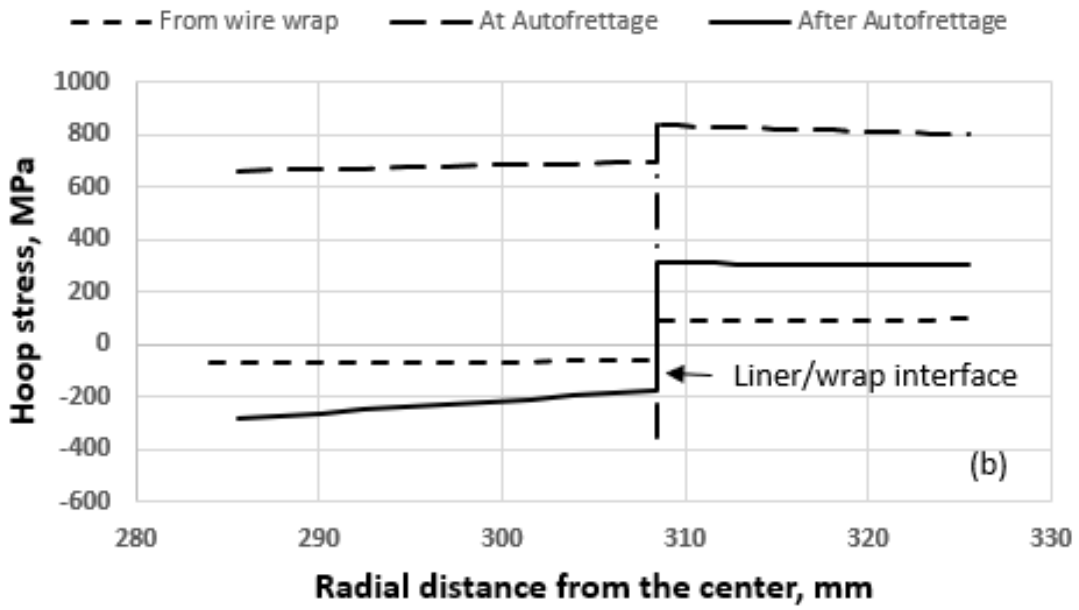


Fig. 2- Hoop stresses in the wall of the liner and the wire jacket after wire winding, at the autofrettage pressure and after release of the autofrettage pressure.



Fig. 3 – The wrapped cylinder ready for autofrettage and hydrotesting.

One of the two liners was wrapped in the presence of an ASME authorized inspector (AI) who certified that the vessel was produced according to specified processes. Subsequently, this cylinder was sent to Authorized Pressure Testing in California for autofrettage and hydrotesting. The autofrettage and hydrotesting was also witnessed by the ASME authorized inspector who has certified the cylinder as an official prototype of the vessel.

This design can now be manufactured and be offered commercially for integration in hydrogen storage facilities. To move the technology to TRL levels of 9 and 10, it is necessary to have these cylinders placed in service. The advantages of this newly developed Type II-S cylinder and a strategy to raise the TRL level to 10 are briefly described below.

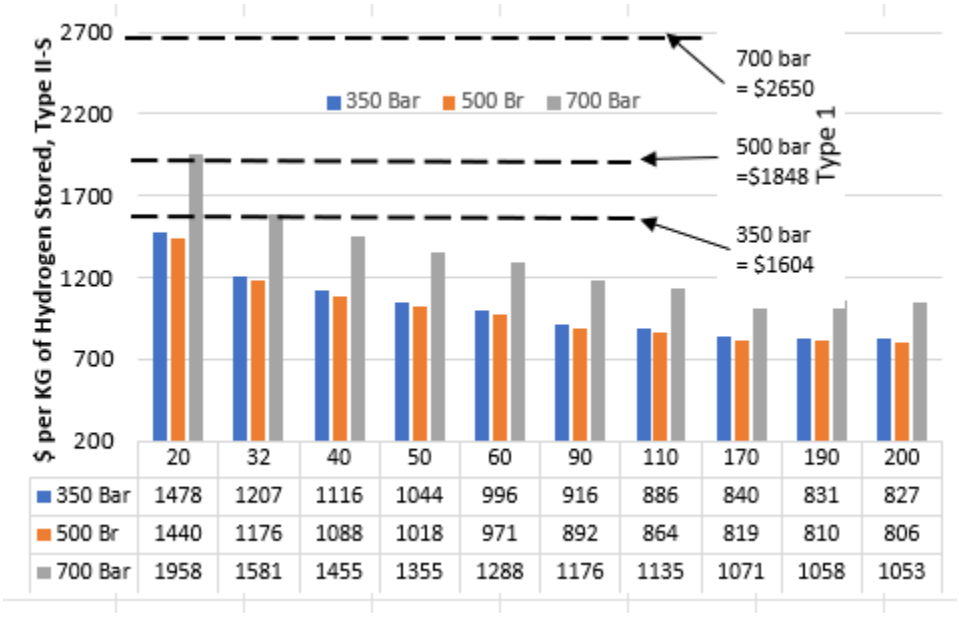
The post-project plan is to commercialize the 350-bar vessel and to seek funding to develop the 500 bar and 700 bar vessels. The primary advantages of the newly developed Type II-S vessels over the currently used Type 1 cylinders in hydrogen storage applications are cost and the cylinder

weight. Both are expressed quantitatively in terms of per KG of hydrogen stored. A single manufacturing line producing Type II-S cylinders working in two-shifts can produce up to 200 cylinders annually. This is called the annual production volume (APV). The manufacturing costs of a Type II-S cylinder and the price of Type I cylinder is shown in Table 1. Also listed in Table 1 are the weights per KG of H2 stored for both Type 1 and Type II-S cylinders. The cost of the 350 bar Type II-S cylinder is 48.5% less than the price of Type 1 cylinder. Further, weight savings of 55% will be realized by switching from Type 1 design to a bulk-produced (APV of 200) Type II-S design. At higher hydrogen pressures, such as 500 and 700 bar, the cost and weight savings are similar. Figure 4 shows a comparison between manufacturing costs of Type II-S and the price of Type 1 cylinders at all three pressures.

Table 1- Cost/price and weight savings in switching from Type 1 to Type II-S cylinders for MOP of 350 bar.

MOP, Bar	Cost/KG of H2 Stored ($)			Cylinder Weight/KG of H2 Stored, KG		
	Type II-S Cost	Type 1 Price	% Savings	Type II-S	Type 1	% Savings
350	827	1604	48.5	90.4	200.5	55.0

WireTough is primarily an R&D company with facilities for prototyping and small-scale manufacturing and does not have the capacity to meet the high demand for cylinders. It is thus best to make the technology available for licensing to large scale manufacturers who can further optimize costs. Therefore, the proposed commercialization strategy is to find licensing partners who can produce the cylinders on a large-scale. The ideal licensee candidates are manufacturers of liners or the Type 1-cylinder manufacturers. The liners used in the construction of these cylinders constitute approximately 50% of the cylinder cost. Figure 5 shows the various elements of the manufacturing costs; these costs are dominated by materials costs and trend similarly for cylinders with pressure ratings of 350, 500, and 700 bar. It is seen that the overall costs expressed as $/KG of H2 stored change only somewhat with increasing maximum operating pressure (MOP). Thus, the choice of MOP can be based on other factors such as compression costs, available space for the storage facility, and the end uses of the hydrogen being stored. It is also important that manufacturing facilities to be globally distributed to serve the needs of their respective regions and thereby minimize shipping costs and delivery times which are better facilitated by the proposed licensing model.



	20	32	40	50	60	90	110	170	190	200
350 Bar	1478	1207	1116	1044	996	916	886	840	831	827
500 Br	1440	1176	1088	1018	971	892	864	819	810	806
700 Bar	1958	1581	1455	1355	1288	1176	1135	1071	1058	1053

Fig. 4 – Cost of hydrogen storage in $ per KG of hydrogen stored using Type II-S cylinders as a function of annual production volume (APV) in a single production line for various MOP values. The cost is compared to the cost of using Type 1 cylinders.